ATLAS HL-LHC Performance Meeting (DP&P session)

AI-Based Energy Reconstruction for the ATLAS Tile Calorimeter at HL-LHC

A calibrated teacher, a sub-100-parameter distilled student, and the path to the FPGA

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Content

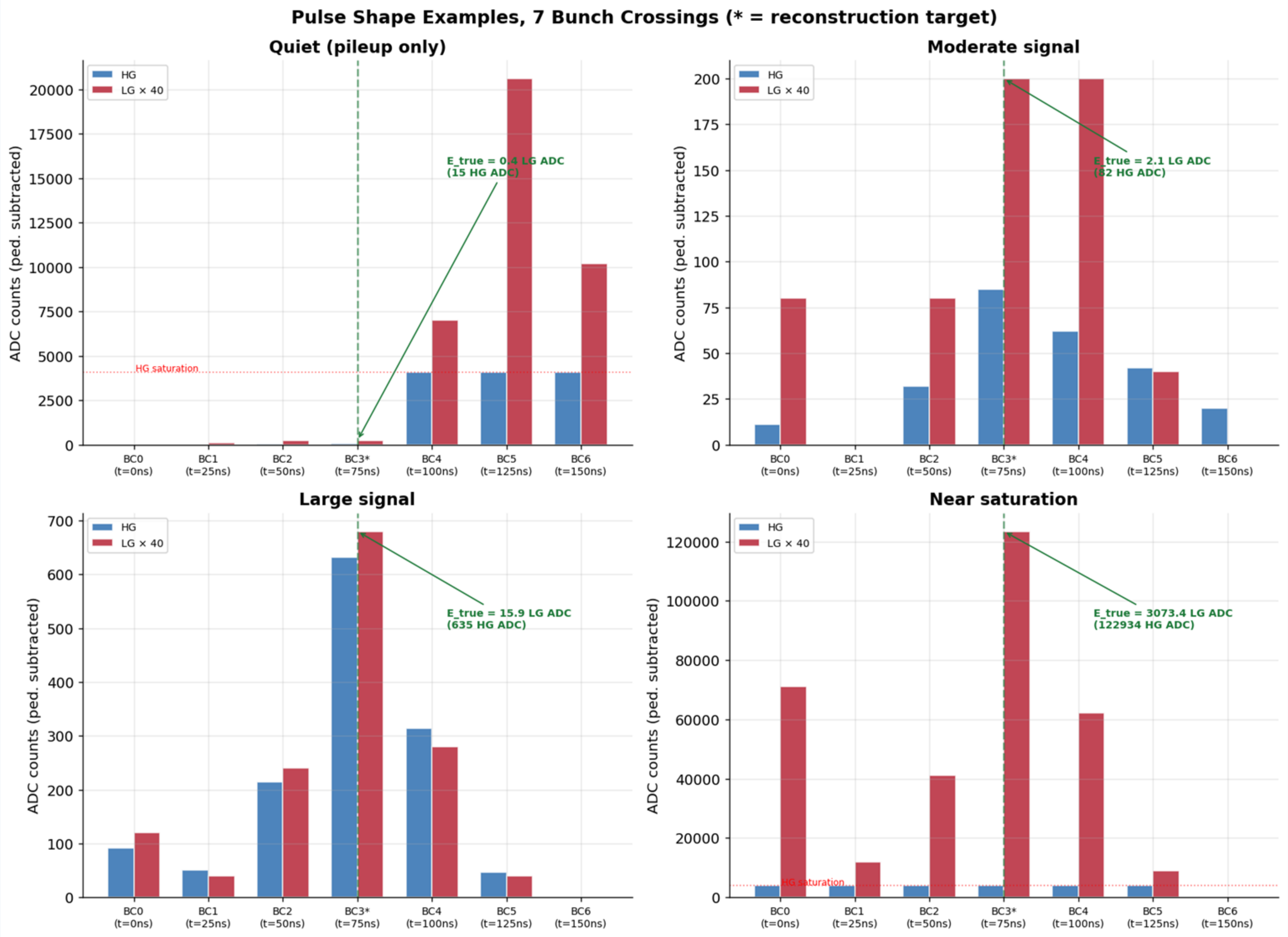
Motivation and Goal

At the HL-LHC, TileCal sees about **200 simultaneous proton-proton collisions** per bunch crossing (pile-up, <mu> = 200).

Energy is reconstructed in real time, per bunch crossing, on the **TilePPr FPGA**.

The legacy **Optimal Filtering (OF)** algorithm degrades under this pile-up because of non-Gaussian out-of-time contributions.

Goal: replace OF with a compact neural network that fits the TilePPr FPGA latency and resource budget. The input is a **9-sample sliding window** (9 bunch crossings, 225 ns of detector time), target is order-100 parameters per channel, for about 77 channels.



pulse-shape example with out-of-time pile-up (HG saturated, LG captures the peak)

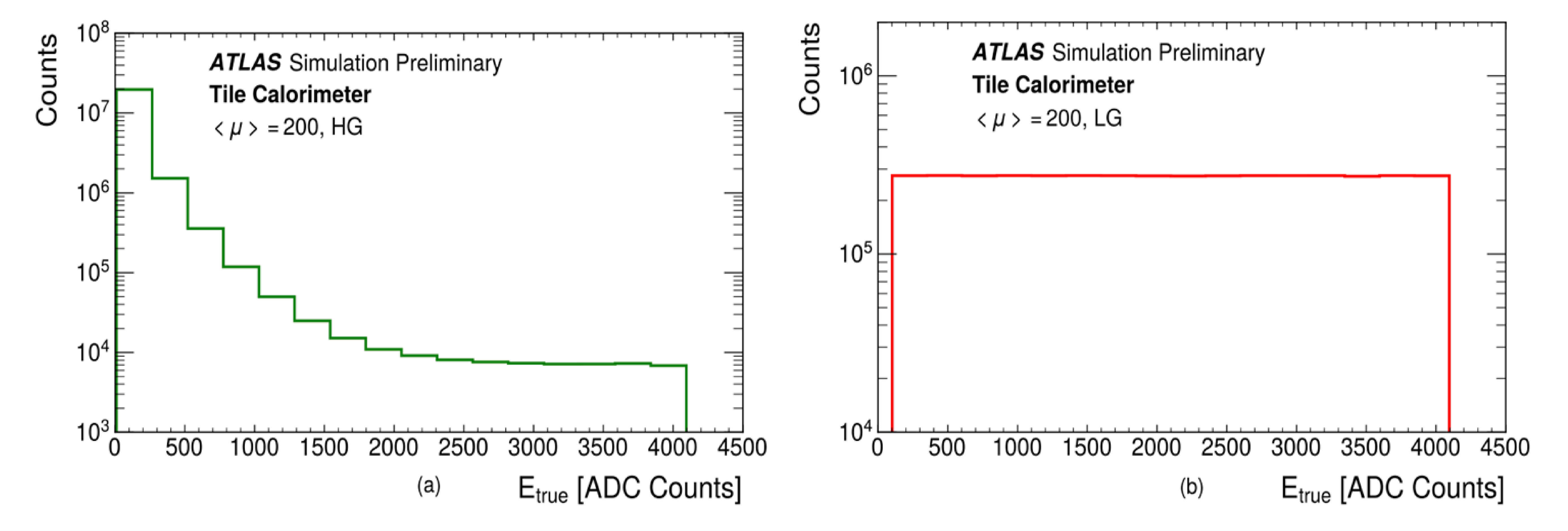
The Data

About **1M consecutive bunch crossings** simulated at <mu> = 200 minimum-bias, with a 5% superimposed flat signal distribution. Long-barrel **A-layer cells A1–A5**.

Each channel is read in two gains, **High Gain (HG)** and **Low Gain (LG)**, a factor of 40 apart, both 12-bit ADCs (0–4095 counts), pedestal about 100 counts (not constant).

Input: sliding window of **9 bunch crossings**. Target: true energy of the central bunch crossing.

About **95% of windows** fall in the lowest-energy bin (pure pile-up), a steep low-energy spike with a flat signal tail.



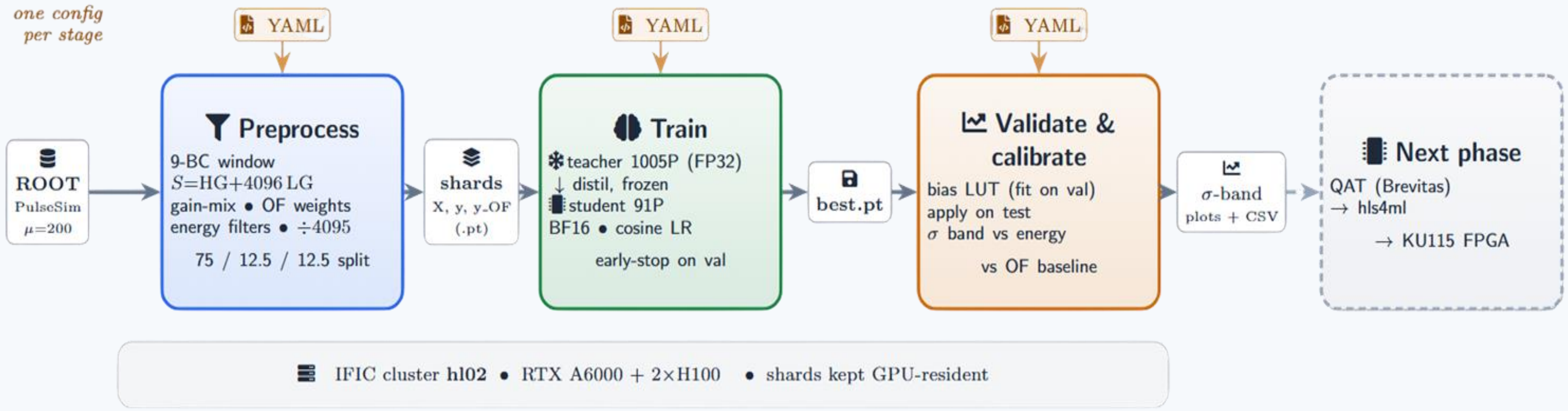
ATLAS-approved public plot. (NOTE: public plot is cell A1, our training set is A1–A5, same mu=200 + 5% flat-injection setup)

Pipeline and Setup

Full three-stage pipeline on the **IFIC GPU cluster (hl02)**: preprocess (ROOT to shards), train, validate and plot. Config-driven, one YAML per stage.

Multi-cell A1–A5 datasets at window size 9, built once and cached.

The training data is kept **GPU-resident**. The A1–A5 epoch dropped to about **2 min 30 s** (down from roughly 10–15 min), bit-for-bit identical to the old path, which made the full set of tests feasible.



three-stage pipeline flow

The Unified Dual-Gain Input

Single concatenated input **S = HG + 4096*LG** (HG in the low 12 bits, LG in the high 12 bits), one target ene_lo. Example, one BC: HG = 820, LG = 20 -> S = 820 + 4096*20 = 82,740. This replaced an earlier two-channel, two-head design.

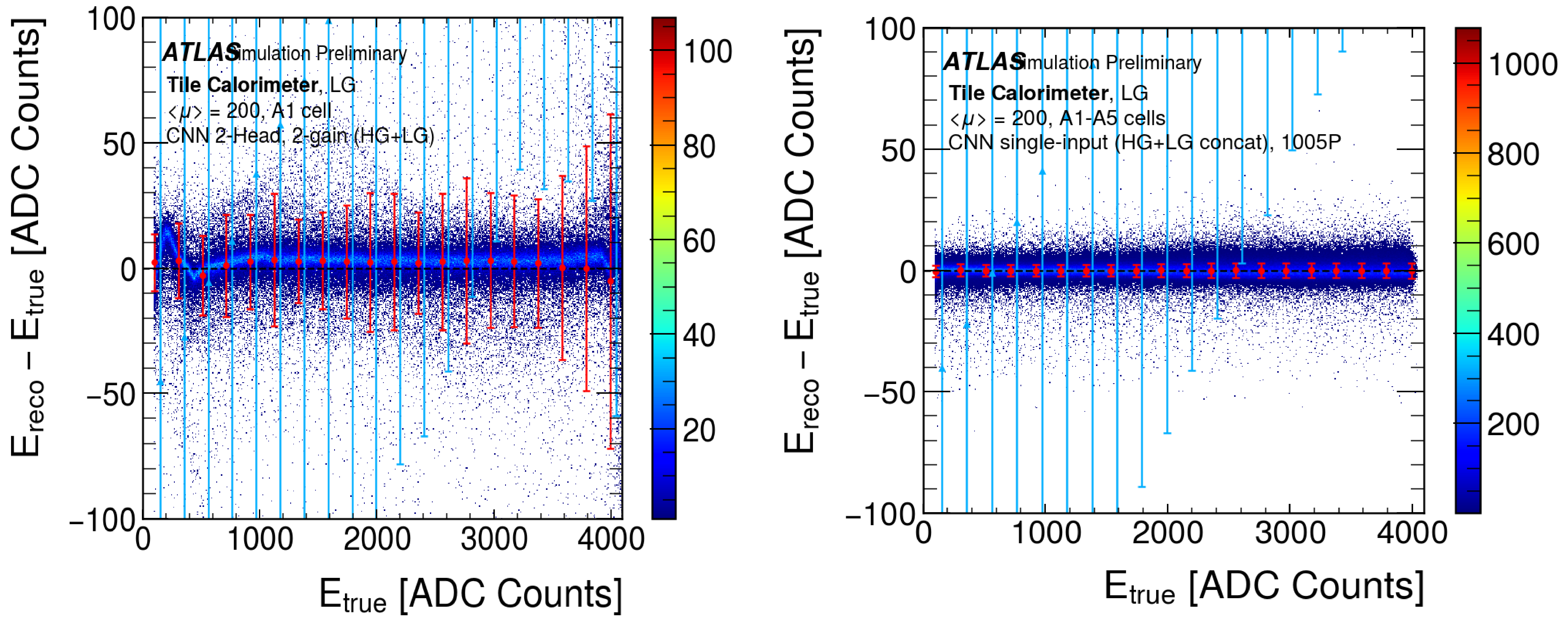
Why it helps?? 1 model takes both gains as a single input and gives one energy output, so no separate per-gain reconstruction or training. The HG is the target x40.

FPGA-friendly activation: **LeakyReLU(0.25)** = a 2ˆ-2 bit-shift (parameter-free, replaces PReLU).

Fixed the low-gain **"S-shape"**: the old two-head design had an energy-dependent LG bias that curved like an "S" across the range; the unified input removes it, the LG mean is now flat at 0 with about 3 LG ADC width across the whole range.

This model is distilled from a **larger, more accurate teacher.**

The Unified Dual-Gain Input



LG band before vs after the unified input (S-shape removed)

The Model: 1D-CNN Architecture

A compact 1D-CNN, two convolutions then a linear read-out:

Conv1d(1->c1, k5) -> LeakyReLU(0.25) -> Conv1d(c1->c2, k3) -> LeakyReLU(0.25) -> Flatten -> Linear(9*c2 -> 1)

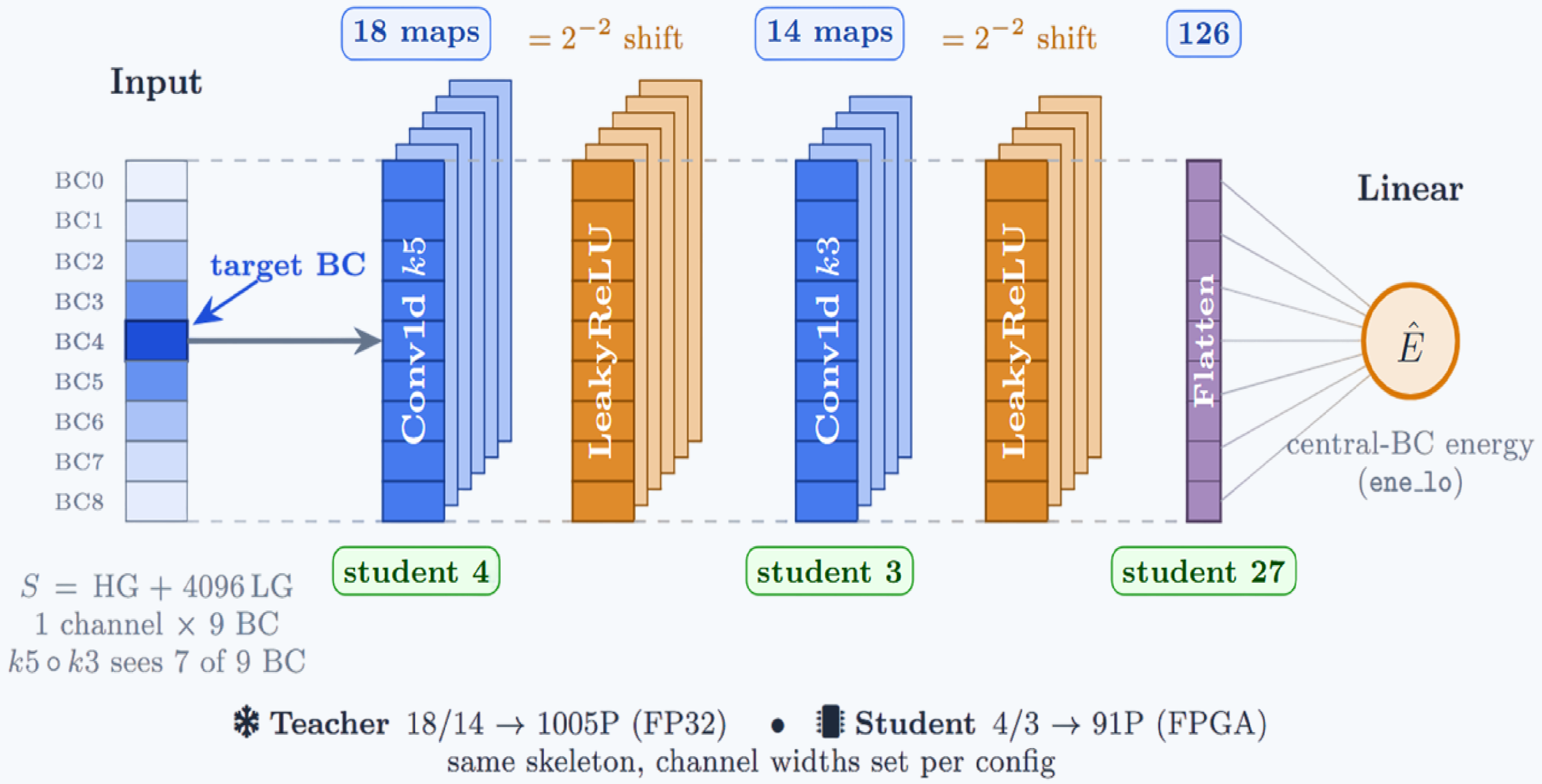
One input channel (concatenated dual-gain), one output (central-BC energy).

Physics-driven kernels: k=5 then k=3 gives a 7-BC receptive field, enough to see out-of-time pile-up at window edges, two k=3 layers would reach only 5 BC.

FPGA-friendly by construction: LeakyReLU(0.25) = 2ˆ-2 bit-shift (no multiply, parameter-free), small integer channel widths, single linear head.

Two sizes: teacher c1/c2 = 18/14 -> **1005 parameters**; FPGA student c1/c2 = 4/3 -> **91 parameters**, reached by distillation.

Training: BFloat16 mixed precision on hl02, ReduceLROnPlateau / cosine LR, early stopping on true-label validation loss.



1D-CNN architecture (single concat input -> two conv layers -> linear, teacher 18/14, student 4/3)

Teacher Design Tests (including dead ends)

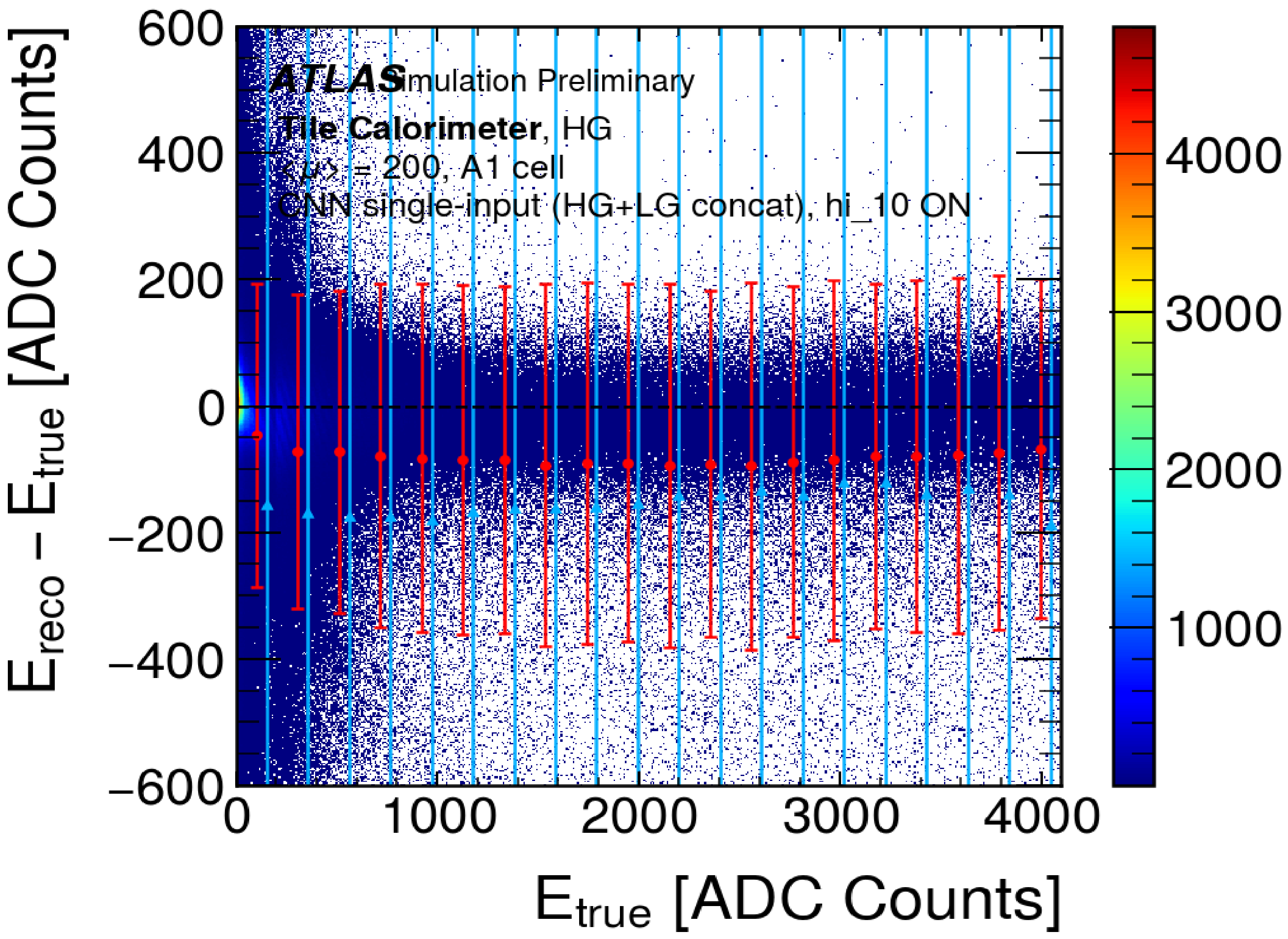
"Drop window if any HG sample < 10 ADC", tested with vs without on a common test set. WITH it the band roughly triples and the mean drifts negative (removes every window containing a quiet BC), so **kept OFF**.

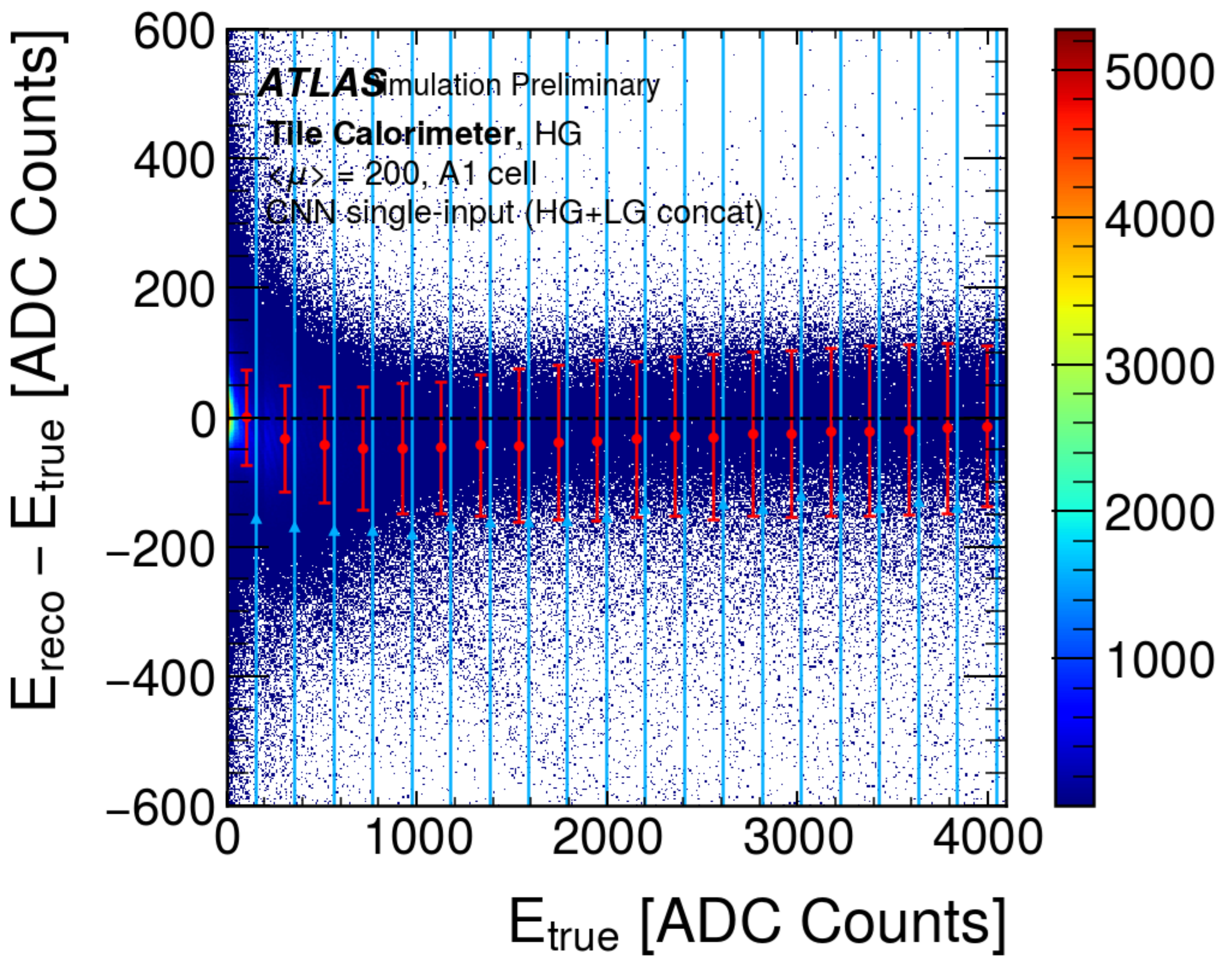
8 vs 9 BC: dropping to 8 BC removes the **LAST BC** (more reconstruction time for the FPGA), but HG is about 5% wider, so we stay on **9 BC**.

Multi-cell A1-A5: resolution matches single-cell A1, so one model covers the A-layer cells.

Dead end: an energy-weighted loss put weight on the already-good high-energy region and made HG slightly worse, so **dropped**.

Teacher Design Tests (including dead ends)





with-cut vs without-cut HG band on the common test

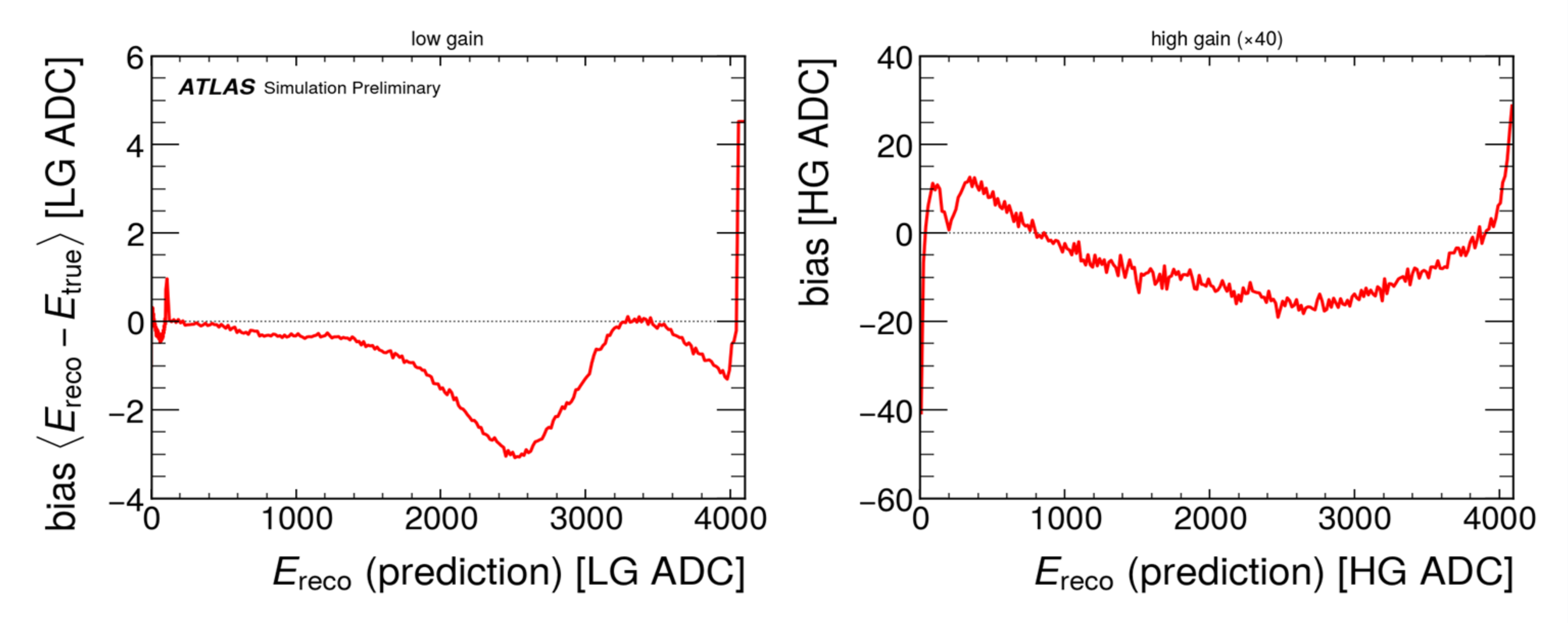
The HG Offset and Calibration

A raw HG offset of about **-20 HG ADC** (positive in the first bin) comes from the single LG target, a tiny target bias of about -0.5 LG ADC amplified ×40. The reference does not see it because it uses separate per-gain models.

Calibration: measure the bias as a function of the prediction (reco energy) on validation, then subtract it on the reco energy. **One curve fixes both gains.**

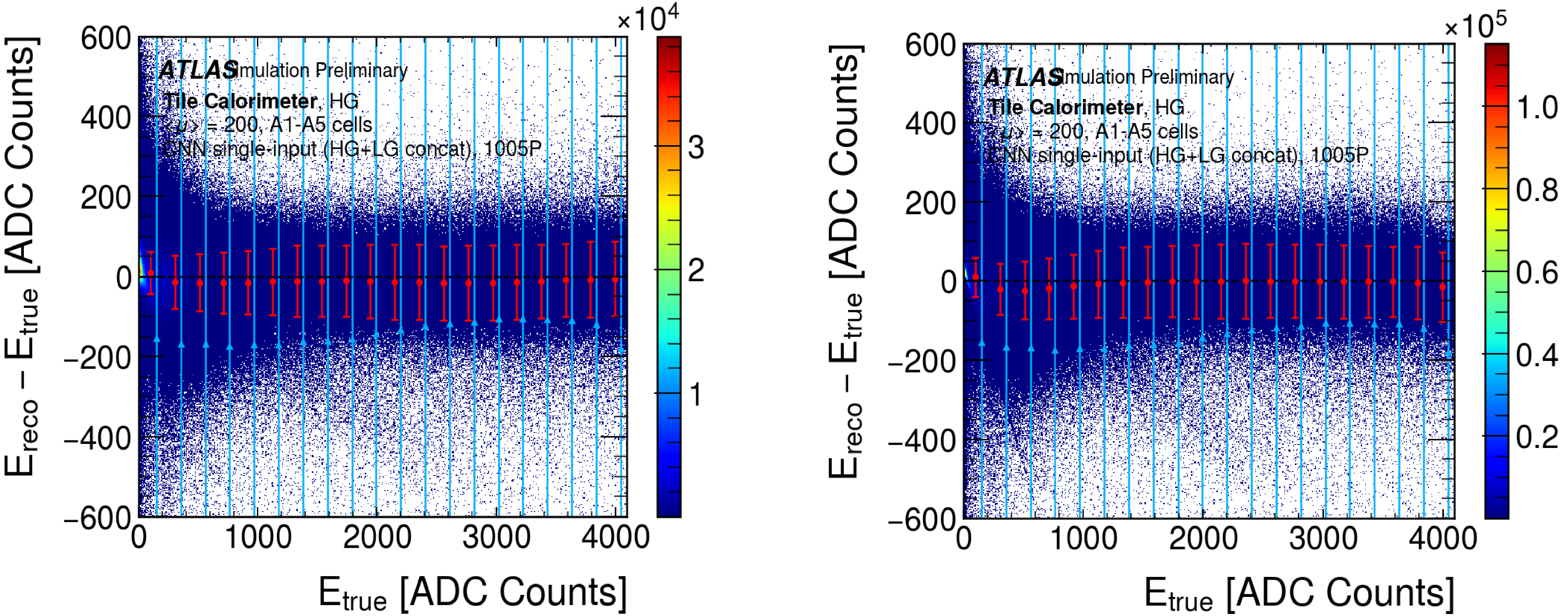
Granularity: HG events sit at low LG energy, so uniform bins gave the HG side far fewer points than LG, we now bin the HG region with 256 points matching the LG side.

The offset is removed. A residual very-low-energy HG dip remains and it is regression-to-the-mean not under-correction i.e the dip does not move as the HG binning goes from 6 to 40 to 256 points, while the band sigma stays about 127 HG ADC. The global sigma is dominated by the low-energy bin, so we report the band as the robust number.



bias curve we subtract, low gain [LG ADC] and high gain x40 [HG ADC]

The HG Offset and Calibration



raw vs calibrated HG band (offset removed)

Parameter Sweep to the 1005P Teacher

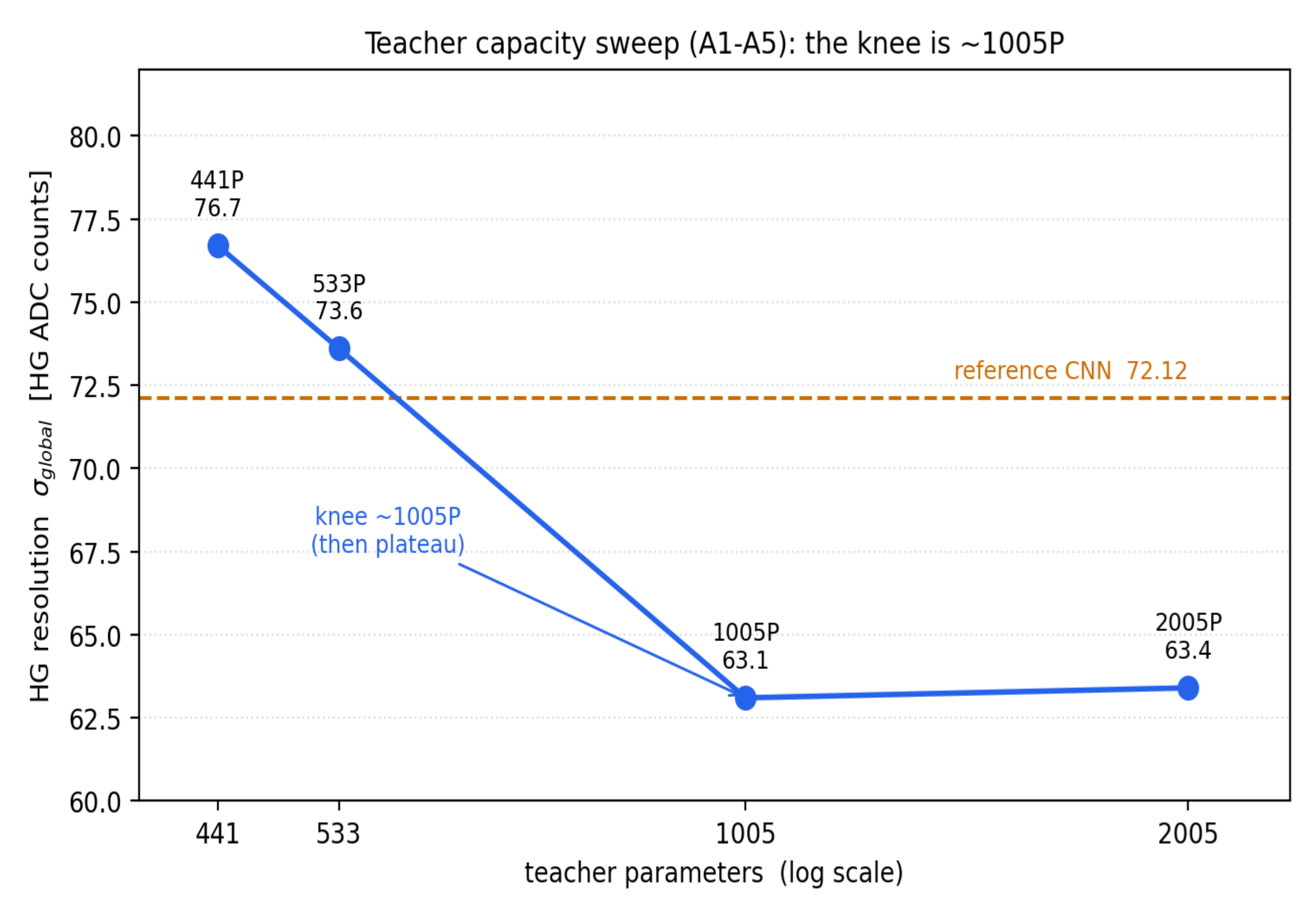
Earlier small models looked like the floor, with A1–A5 data and the speedup, larger models became testable.

HG global sigma vs size: about:–

533P -> 73, **1005P -> 63**, **2005P -> 63**.

At about **1005 parameters** HG tightened in every energy bin and then plateaued at about 2000, so the knee is near 1000 parameters. LG stayed about 2.7 LG ADC throughout.

Reading: extra parameters help once extra input data (more cells) is added. The 1005P CNN is the teacher.



HG global sigma vs parameter count (441/533/1005/2005)

Teacher Results (1005P, A1-A5, calibrated)

Resolution: "global" = std over all events (event-weighted), "band" = mean of per-energy-bin sigma (energy-flat).

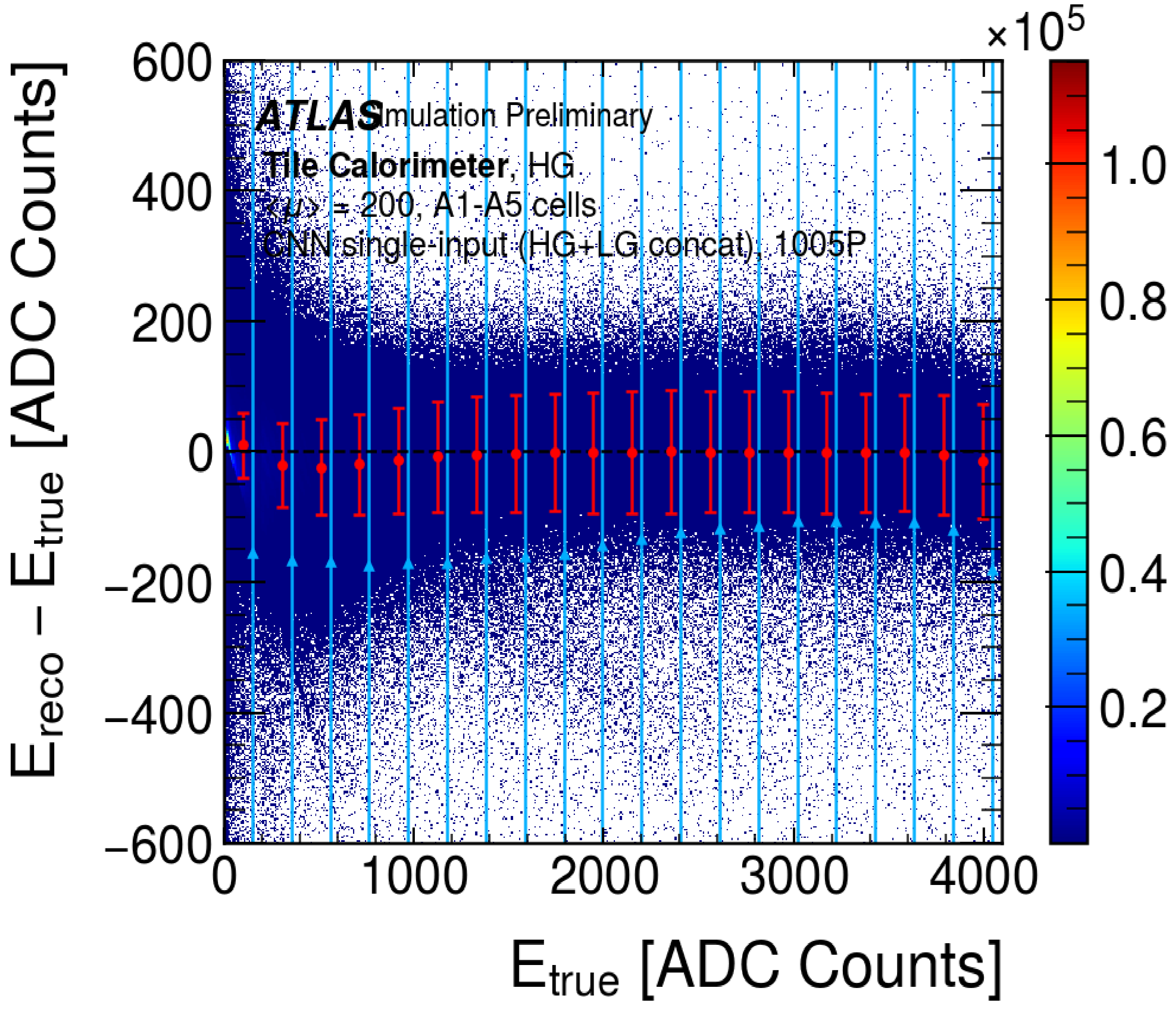
Architecture: Conv1d(1->18, k5) -> Conv1d(18->14, k3) -> Linear(126->1), LeakyReLU(0.25), 1005 parameters. MR: ificuw/tilecal!16 (merged).

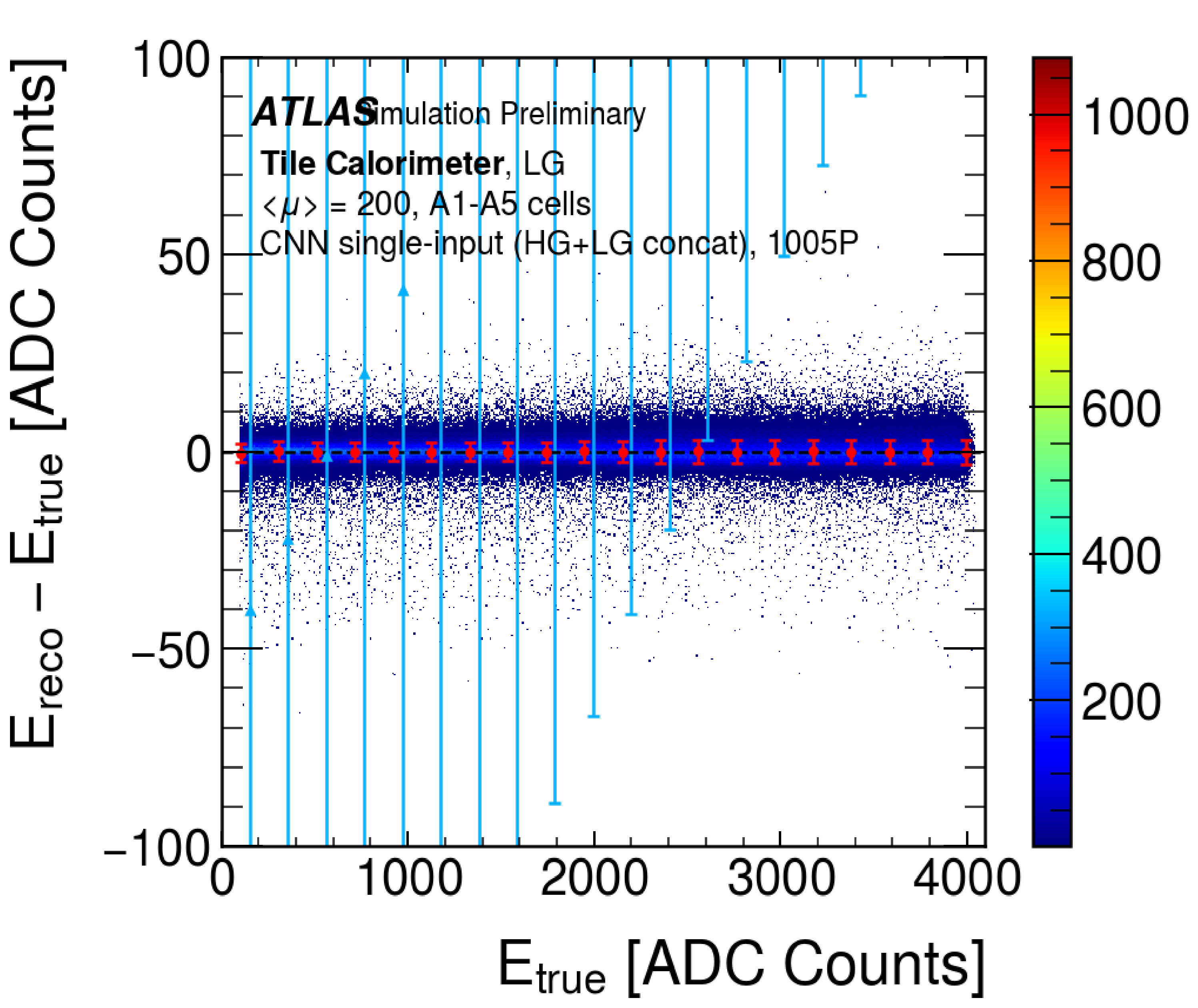
HG band: about +/-53 HG ADC at low energy widening to about +/-93 HG ADC toward the HG/LG boundary, mean flat after calibration, linearity slope about 1.00, R^2 about 0.99.

Model	HG sigma global	HG sigma band	LG sigma
Teacher (1005P, ours)	63.0	85.8	2.70
IFIC ref. CNN (147P, HG-only)	72.12	-	8.36
Optimal Filtering (context)	979.2	890.6	244.9

256-bin HG calibration tightens the global sigma to about 61.6 HG ADC, the band is unchanged, so we quote the band.

Teacher Results (1005P, A1–A5, calibrated)





teacher HG and LG band plots side by side

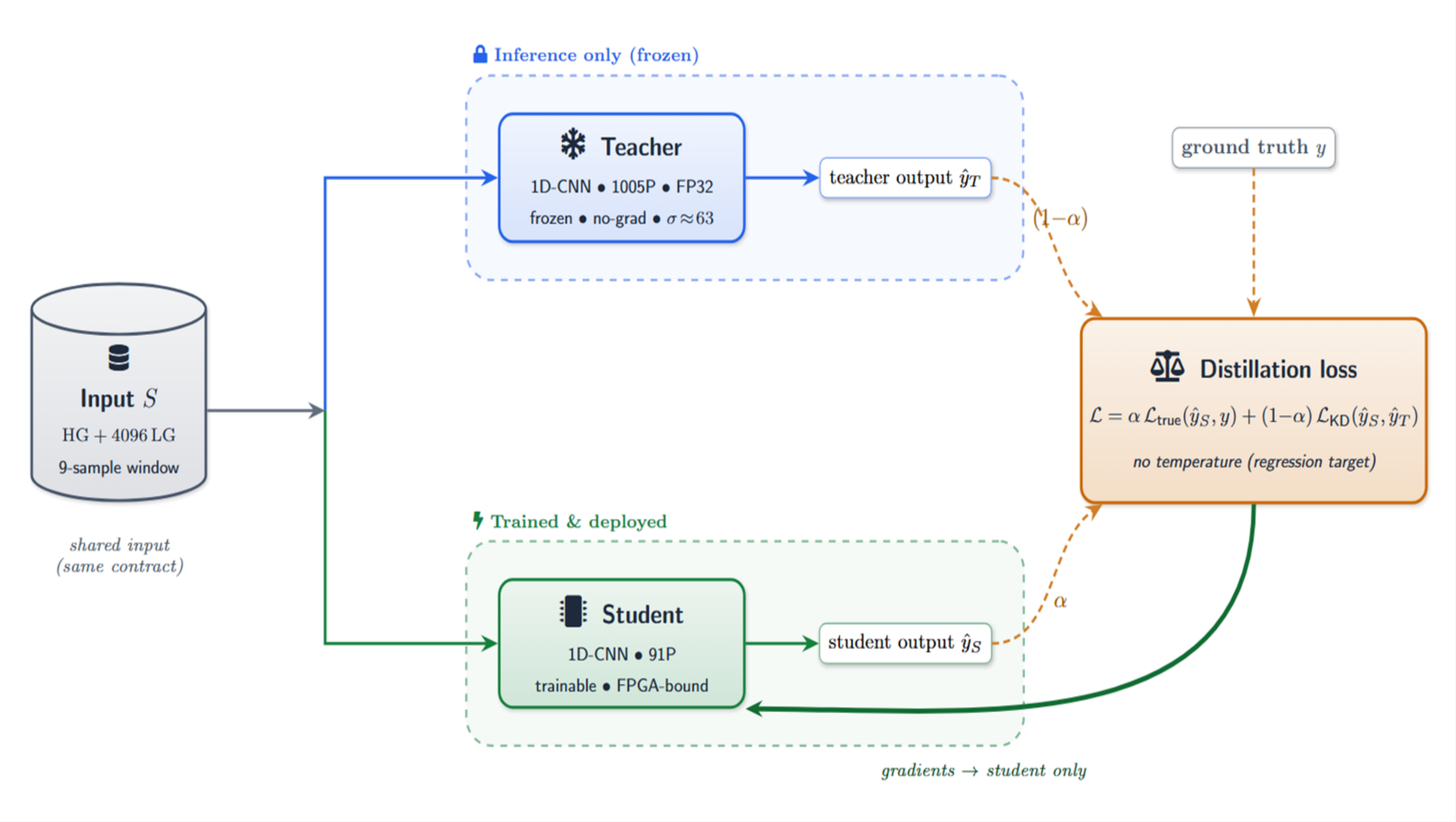
Knowledge Distillation: Framing and Setup

The teacher's HG error no longer improves with more parameter, so distillation here is **variance management**, not accuracy transfer, copying the teacher too literally teaches the student to reproduce noise.

Loss: L = alpha * L_true(student, truth) + (1 – alpha) * L_KD(student, teacher). No temperature (classification construct, this is single-value regression).

The student shares the teacher's exact **input, output, and calibration**. Students trained are 91 and 99 parameter 1D-CNNs.

The KD hook is **additive**: load the frozen teacher, one extra forward pass per batch, add the teacher term. Selection and early-stopping stay on the true-label loss.



teacher -> student distillation (frozen teacher, weighted loss)

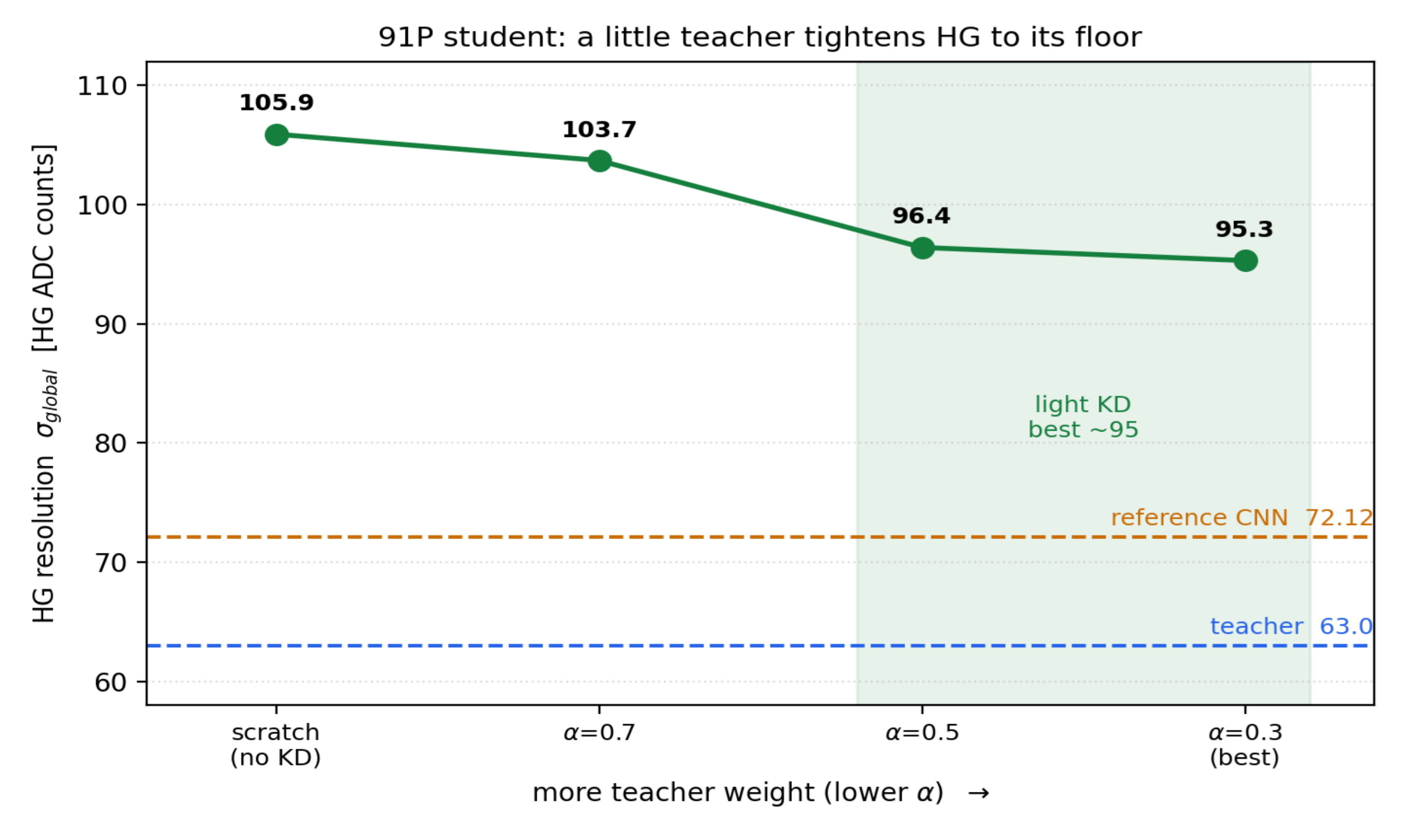
Distillation Results (best student, 91P)

Sweeping KD weight at 91P tightens HG to a floor, from-scratch ~106 -> alpha=0.5 ~96 -> **alpha=0.3 ~95 (best)**. Pushing harder with scale-fixed teacher-matching (99P) overshoots to ~116, worse than from-scratch, over-copies the teacher's noise.

Best student: KD alpha=0.3, 91 parameters. Conv1d(1->4, k5) -> Conv1d(4->3, k3) -> Linear(27->1), LeakyReLU(0.25). The 91P student does **BOTH gains in one model**. On LG it beats both IFIC references (3.43 vs 10.75 MLP, 3.43 vs 8.36 CNN). On HG ~95 ADC (global), comparable to MLP (99.76) and behind CNN (72.12), at the sub-100P capacity floor.

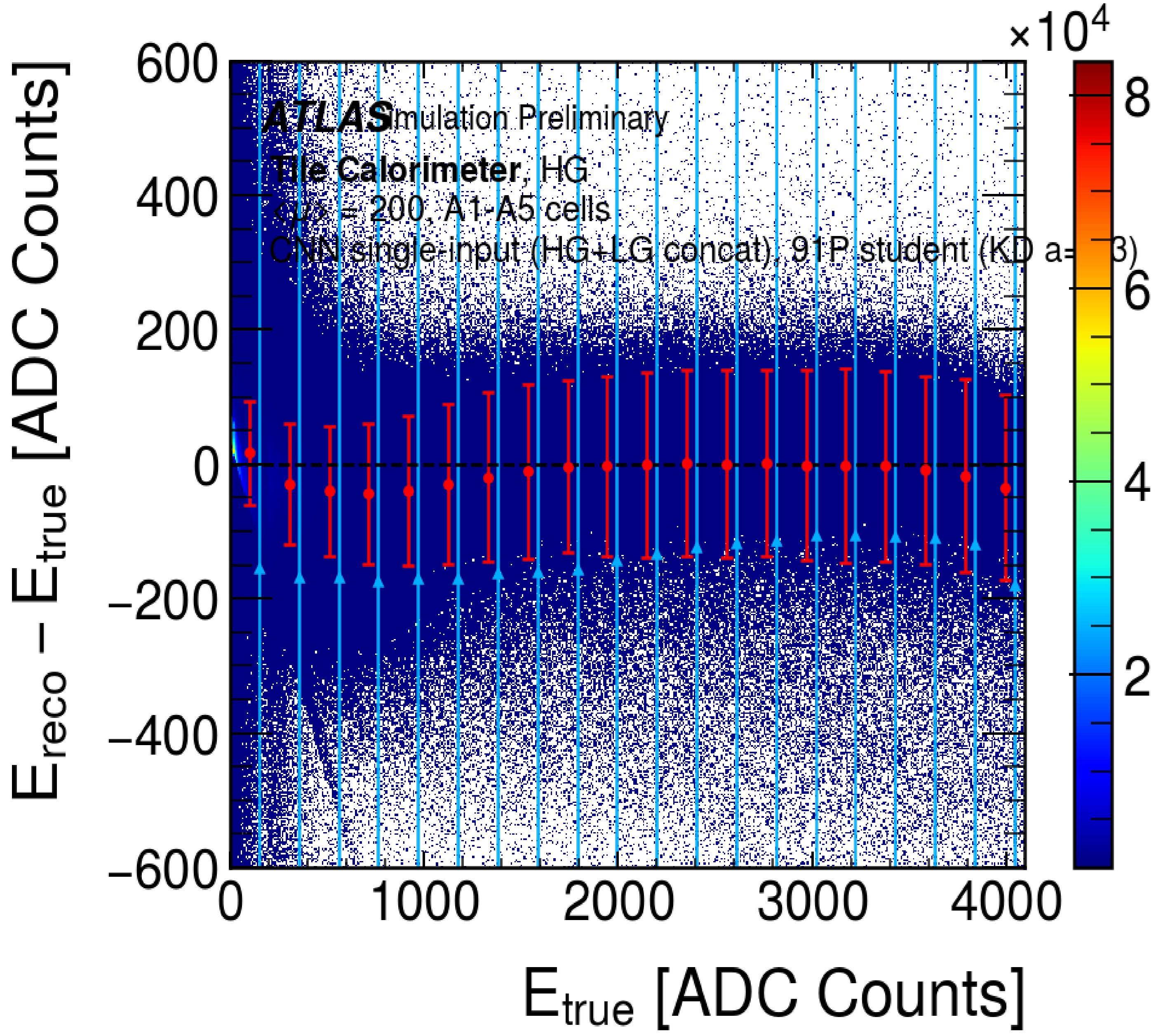
Run	params	HG sigma global	HG sigma band	LG sigma
From-scratch (no teacher)	91	105.9	164.8	4.72
KD alpha=0.3 (best)	**91**	**95.3**	**127.0**	**3.43**
Teacher (ours)	1005	63.0	85.8	2.70
IFIC ref. CNN (HG-only)	147	72.12	-	8.36
IFIC ref. MLP (HG-only)	148	99.76	-	10.75

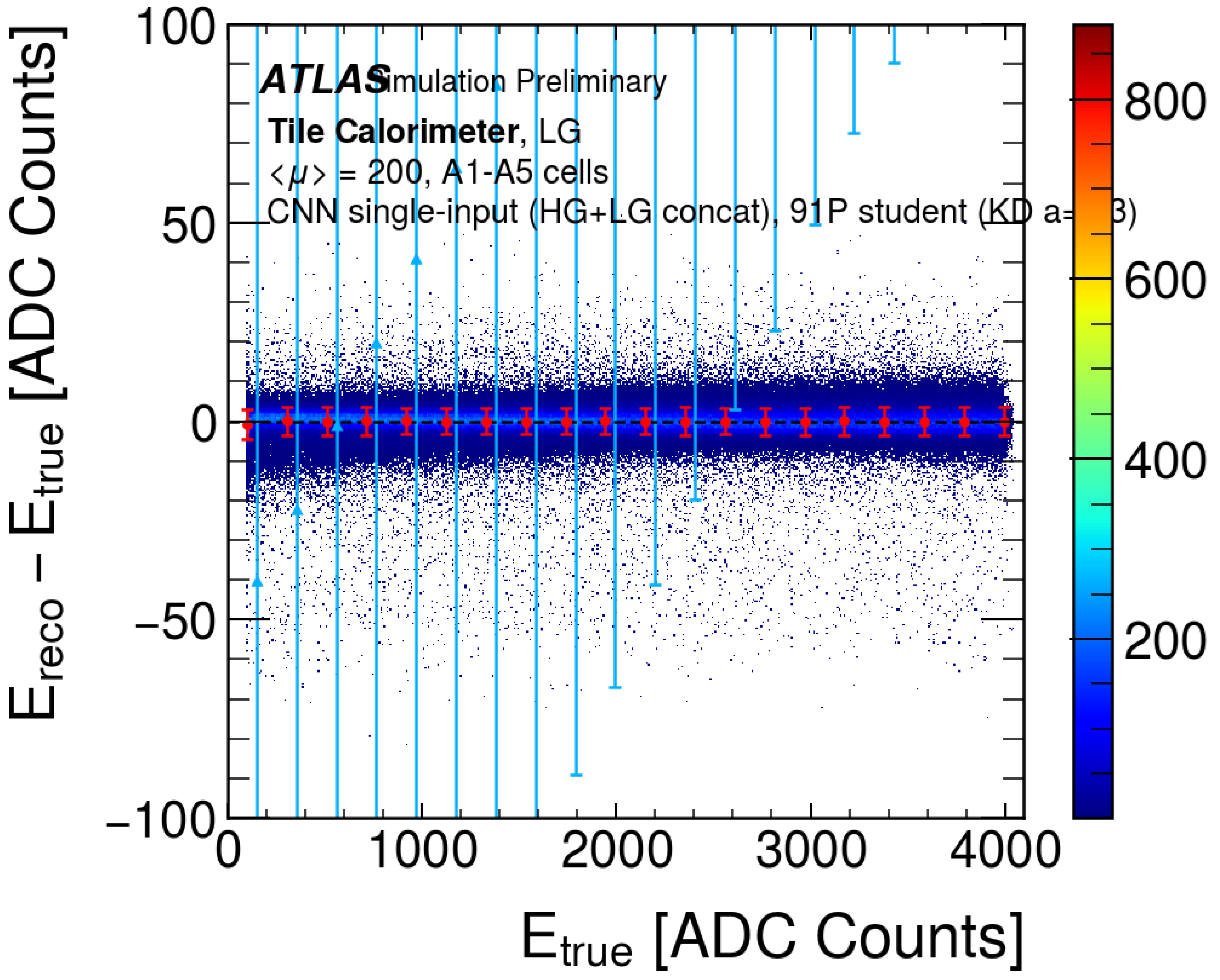
256-bin HG calibration tightens the global sigma to about 91.1 HG ADC, the band is unchanged so, we quote the band.



HG sigma vs KD alpha at 91P

Distillation Results (best student, 91P)





best-student (KD alpha=0.3) HG and LG band plots, CALIBRATED

Summary and Next Steps

Unified input

one concatenated dual-gain sample (S = HG + 4096*LG), one model, one energy output, no separate per-gain reconstruction.

Teacher

A calibrated 1005-parameter CNN, merged (ificuw/tilecal!16). HG comparable to the reference, LG clearly better (about 2.5x).

Student

A working sub-100-parameter distilled model (91P), same input/output and calibration as the teacher. Beats the reference on LG, on HG it reaches about 95 HG ADC at the sub-100-parameter limit.

Next

Quantization: Brevitas QAT from the float model toward about 8-bit, then hls4ml to the TilePPr FPGA (XCKU115-2FLVA1517E), watching the float-to-int drop. MR ificuw/tilecal!17 has the distillation pipeline.

Later (early July): pole-zero / baseline correction, and the Athena offline side.

Open (ML side)

Other cells and layers (BC, D, and the E gap scintillators), pile-up dependence, and timing as a secondary target.

THANK YOU 🙏

Questions?

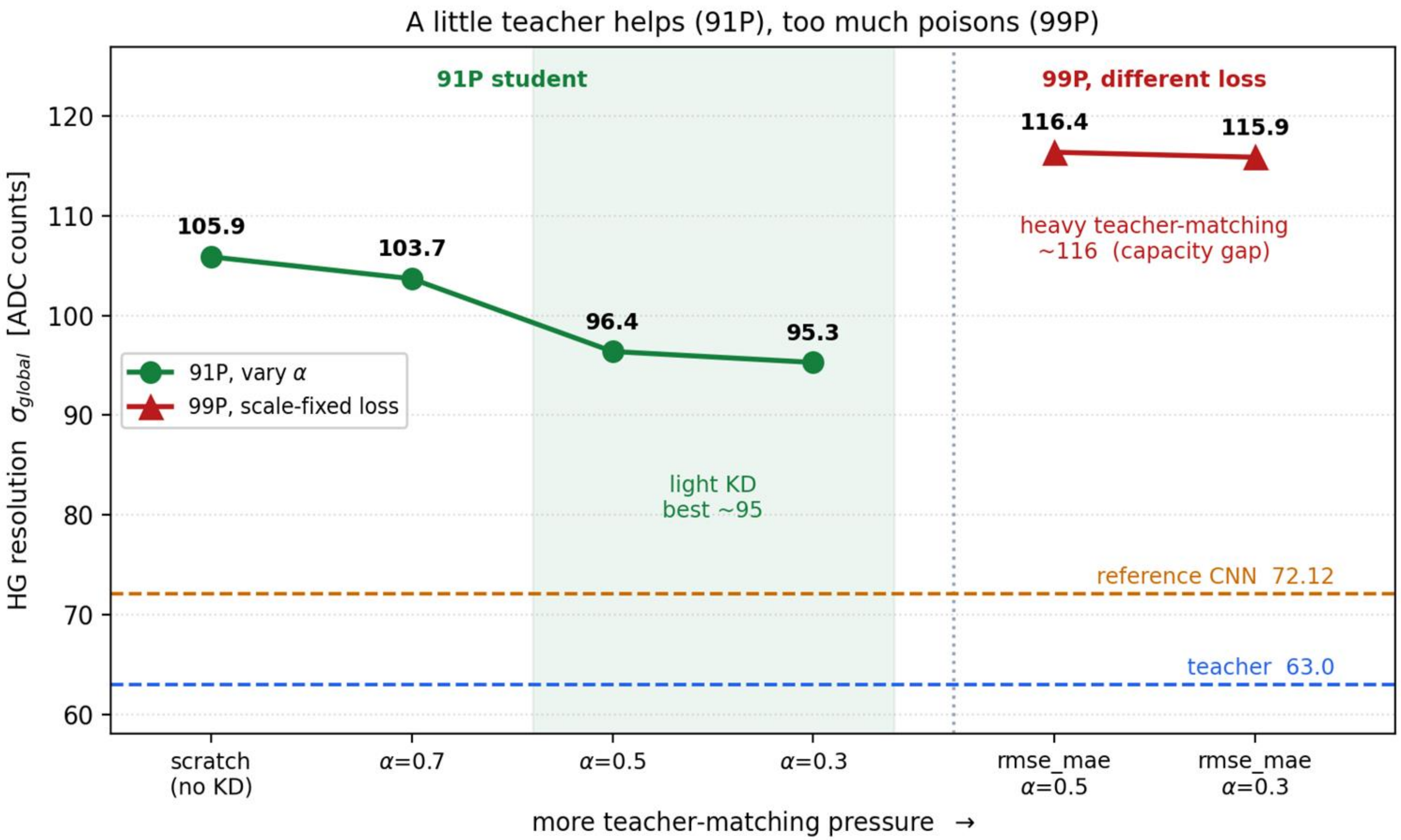
BACKUP

Backup: Full sub-100P student matrix

Every student run, calibrated, on the common A1-A5 test set. HG in ADC counts. Best is KD alpha=0.3 (91P). Heavier teacher-matching (scale-fixed RMSE+MAE) and relative-resolution loss are the two clear dead ends. The band is seed-noisy, the two 106P seeds give 125.7 and 139.6, a -14 ADC swing.

Run	params	distillation	HG sig global	HG sig band	LG sig
From-scratch (no teacher)	91	none	105.9	164.8	4.72
KD alpha=0.3 (best)	**91**	**light, MSE**	**95.3**	**127.0**	**3.43**
KD alpha=0.7	91	light, MSE	103.7	148.3	4.14
KD alpha=0.5	91	light, MSE	96.4	128.5	3.89
KD alpha=0.3	99	light, MSE	106.0	149.6	4.84
KD alpha=0.3	106	light, MSE	98.1	125.7	3.76
KD alpha=0.3, seed 2	106	light, MSE	97.7	139.6	3.77
KD alpha=0.1, teacher 533P	99	MSE	104.8	138.4	4.17
KD alpha=0.1, teacher 1005P	99	MSE	96.6	148.2	4.50
KD alpha=0.3, scale-fixed	99	heavy, RMSE+MAE	115.9	194.4	5.90
k-teacher (5 teachers)	91	MSE	102.3	145.2	3.97
intermediate-layer (feature)	91	MSE + hint	105.8	152.6	4.71
teacher-assistant (441->91P)	91	MSE	101.3	134.1	4.54
relative-resolution loss	91	none, rel loss	109.8	212.6	7.50
-- anchors --					
Teacher (ours)	1005	-	63.0	85.8	2.70

Backup: Full sub-100P student matrix

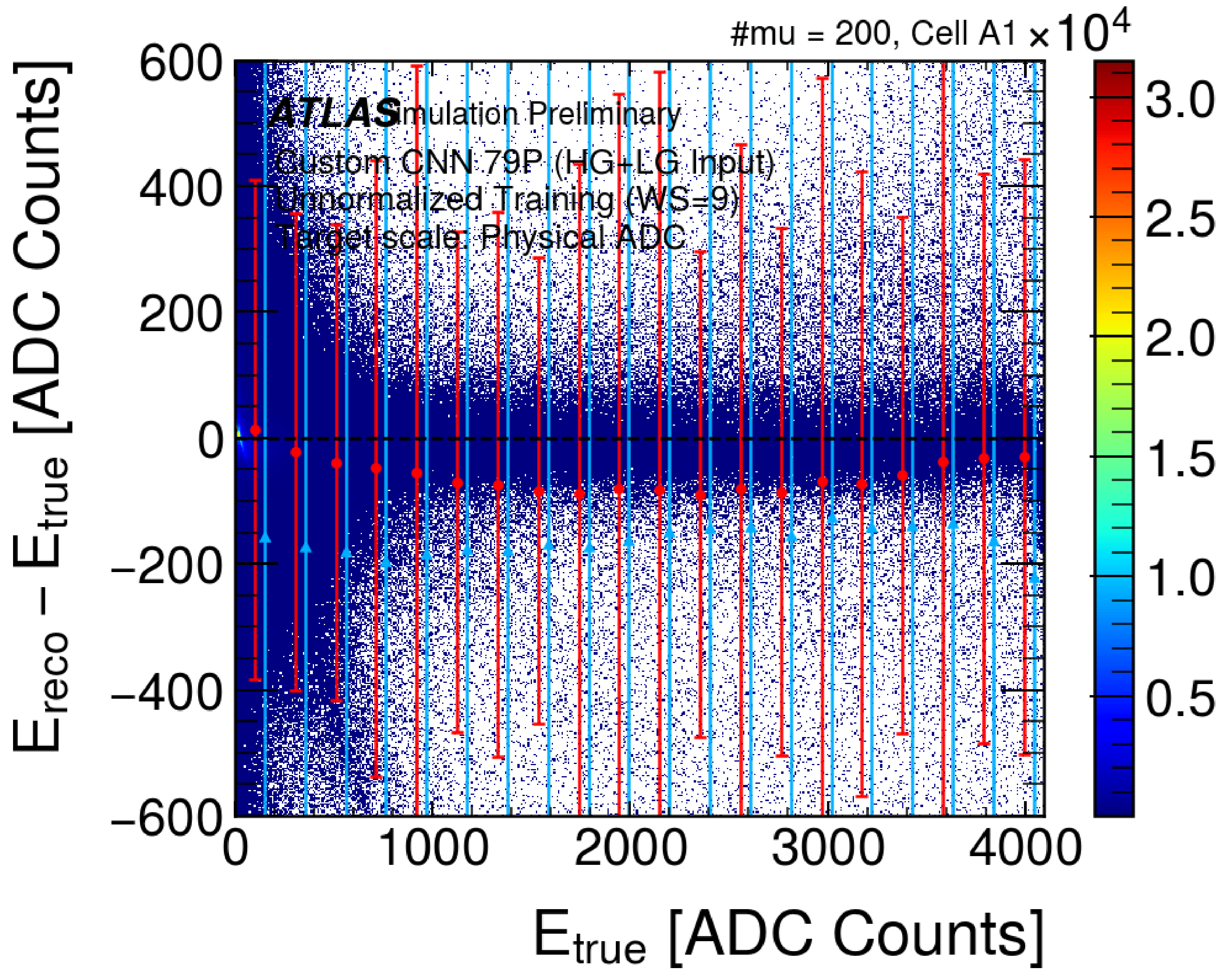


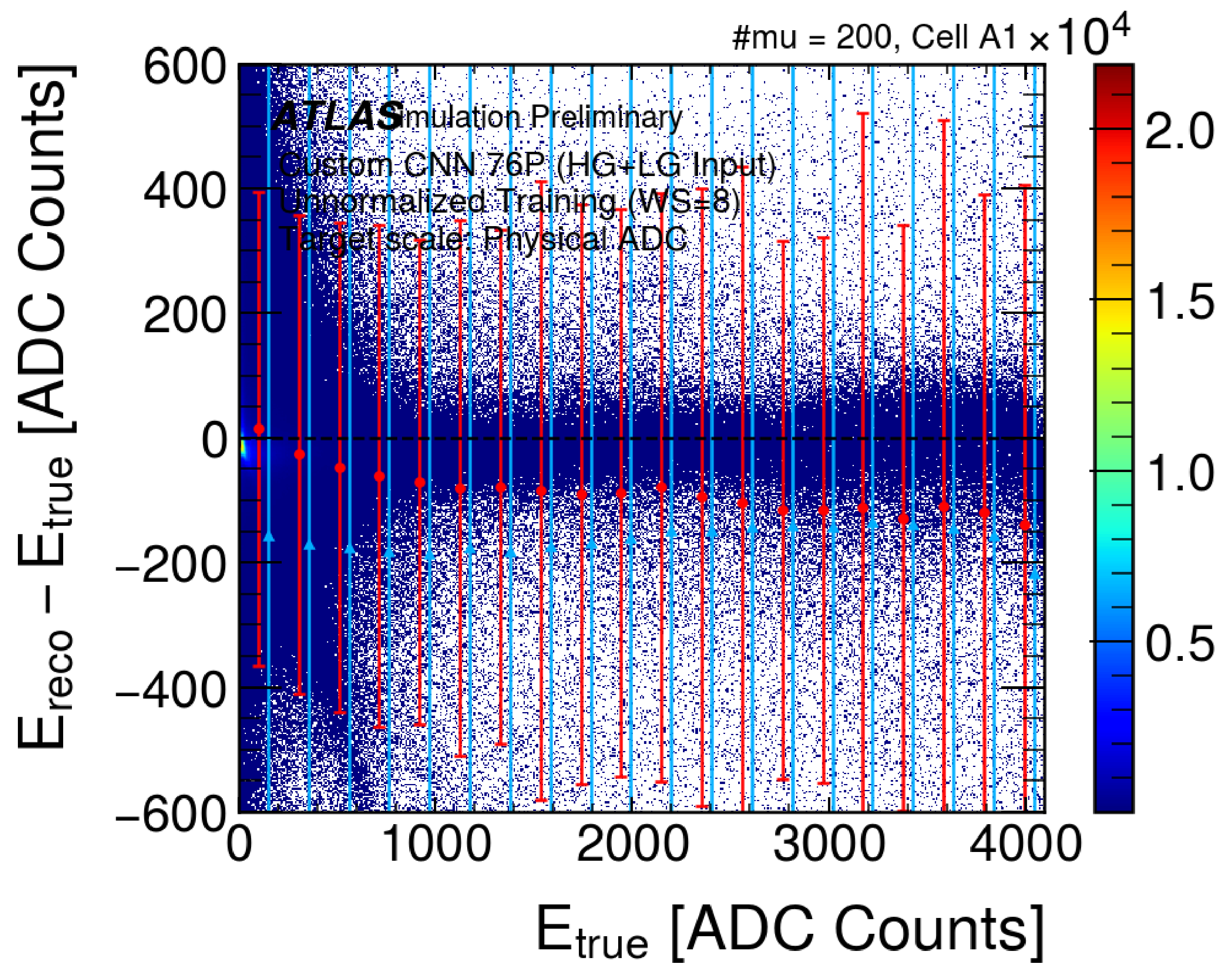
Backup: Dead end – unnormalised target and two-head design

Tested training on **raw ADC with NO target normalisation**, for window size 9 and window size 8. Result? clearly worse. HG R^2 about 0.69–0.70, a large negative bias, and the error sigma "explodes" (raw-ADC scale spread -> gradient instability). **Kept the target normalisation.**

The earlier **two-channel / two-head model** (separate HG and LG heads) was replaced by the single concatenated input **S = HG + 4096*LG**, which fixed the LG S-shape and simplified the network.

Backup: Dead end – unnormalised target and two-head design





unnormalised WS9/WS8 HG absolute-error plots (wide sigma, negative bias)

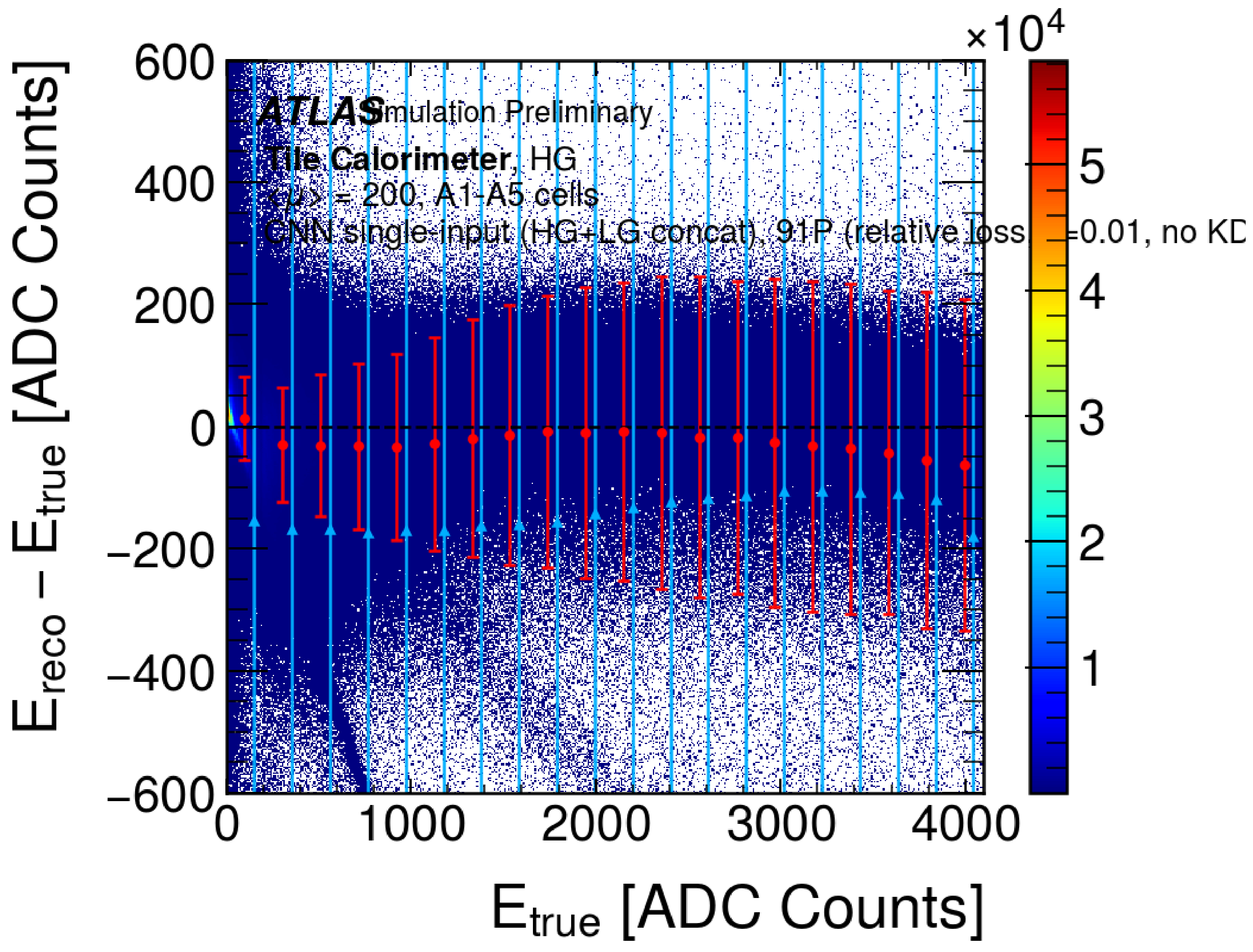
Backup: Dead end – relative-resolution loss

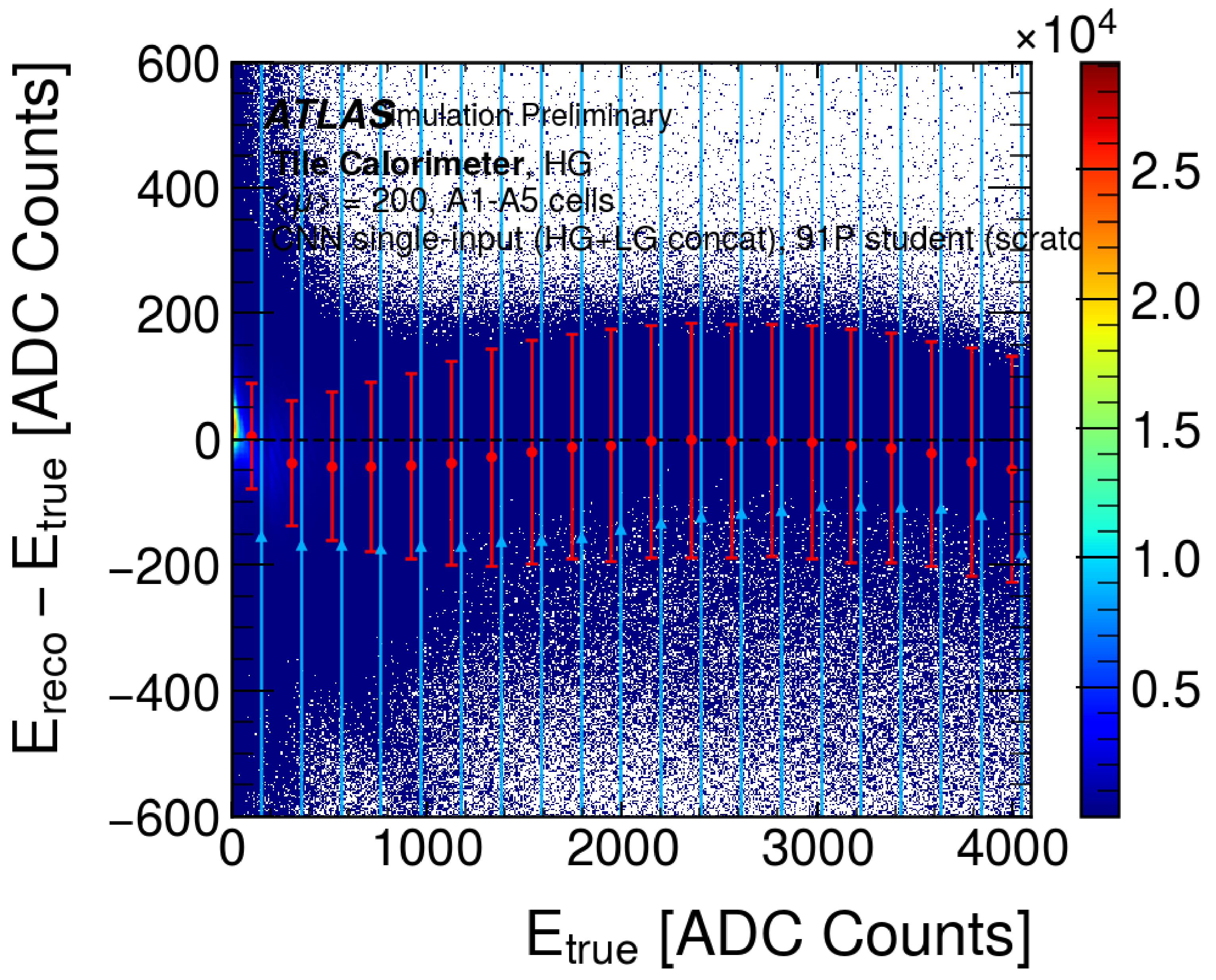
Hypothesis: re-weight the loss toward low energy (fractional resolution sigma(E)/E) to tighten the HG band where it is widest.

Result: worse, not better. HG global about 110 and the per-bin band blew up (bin-mean about 213 vs about 165 from-scratch), LG also degraded.

Why: a relative or absolute-error loss optimises the median, while sigma is set by the mean, and it de-weights high energy, so those bins widened. The loss is not the lever, it reconfirms the **low-energy HG gap is capacity-bound**.

Backup: Dead end – relative-resolution loss





relative-loss HG band (much wider) next to the from-scratch band

Backup: Calibration detail

The bias is computed as a function of **reco energy** (the prediction), not true energy, because true energy is not available at reconstruction time.

Uniform (gentle) bins shift the mean and leave the per-bin sigma unchanged.

A **finer / quantile-binned version** artificially shrank the dominant HG bin sigma (about 14%), so it was rejected as not a real resolution gain.

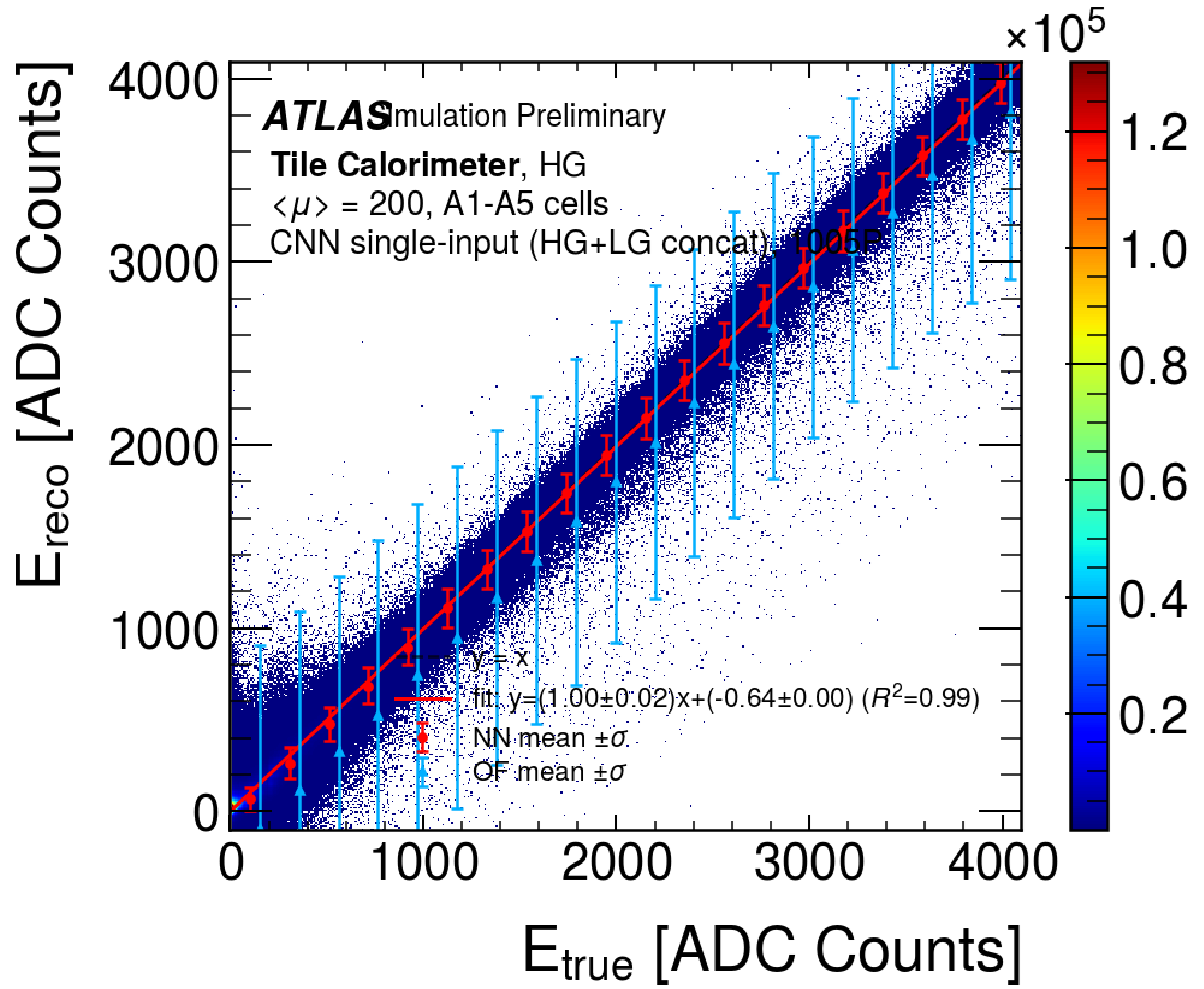
Backup: Linearity and fractional resolution (teacher vs student)

Linearity: E_reco vs E_true sits on the diagonal for both models. Teacher slope about 0.99, R^2 about 0.99, the 91P student stays linear too, so distillation kept it on the diagonal rather than just adding noise.

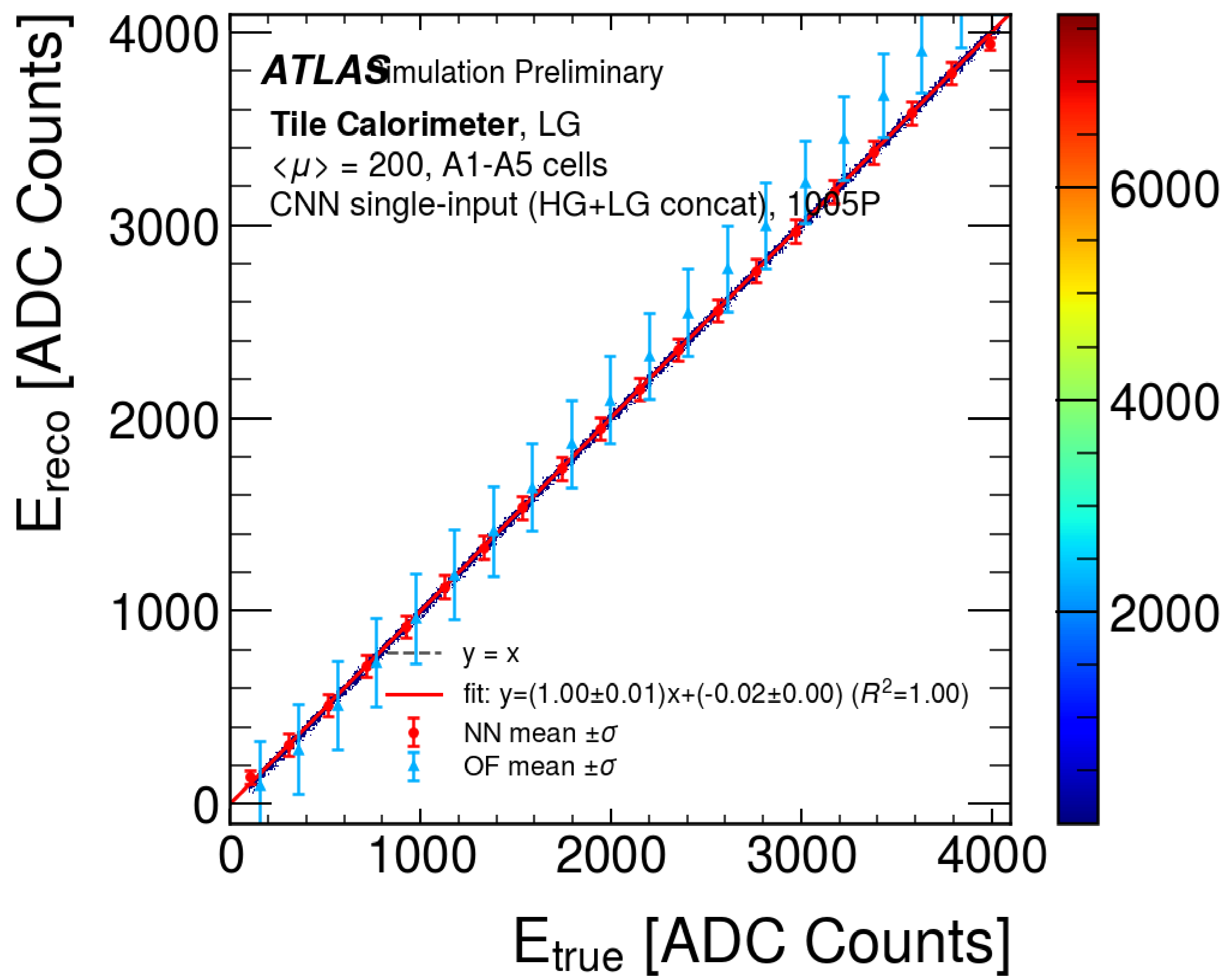
Fractional resolution sigma(E)/E falls with energy and is large only in the lowest-energy bin (pure pile-up, about 95% of events). The student sigma/E is wider than the teacher, the sub-100P capacity gap.

LG is near-perfect for both, HG is where teacher and student differ.

Backup: Linearity and fractional resolution (teacher vs student)

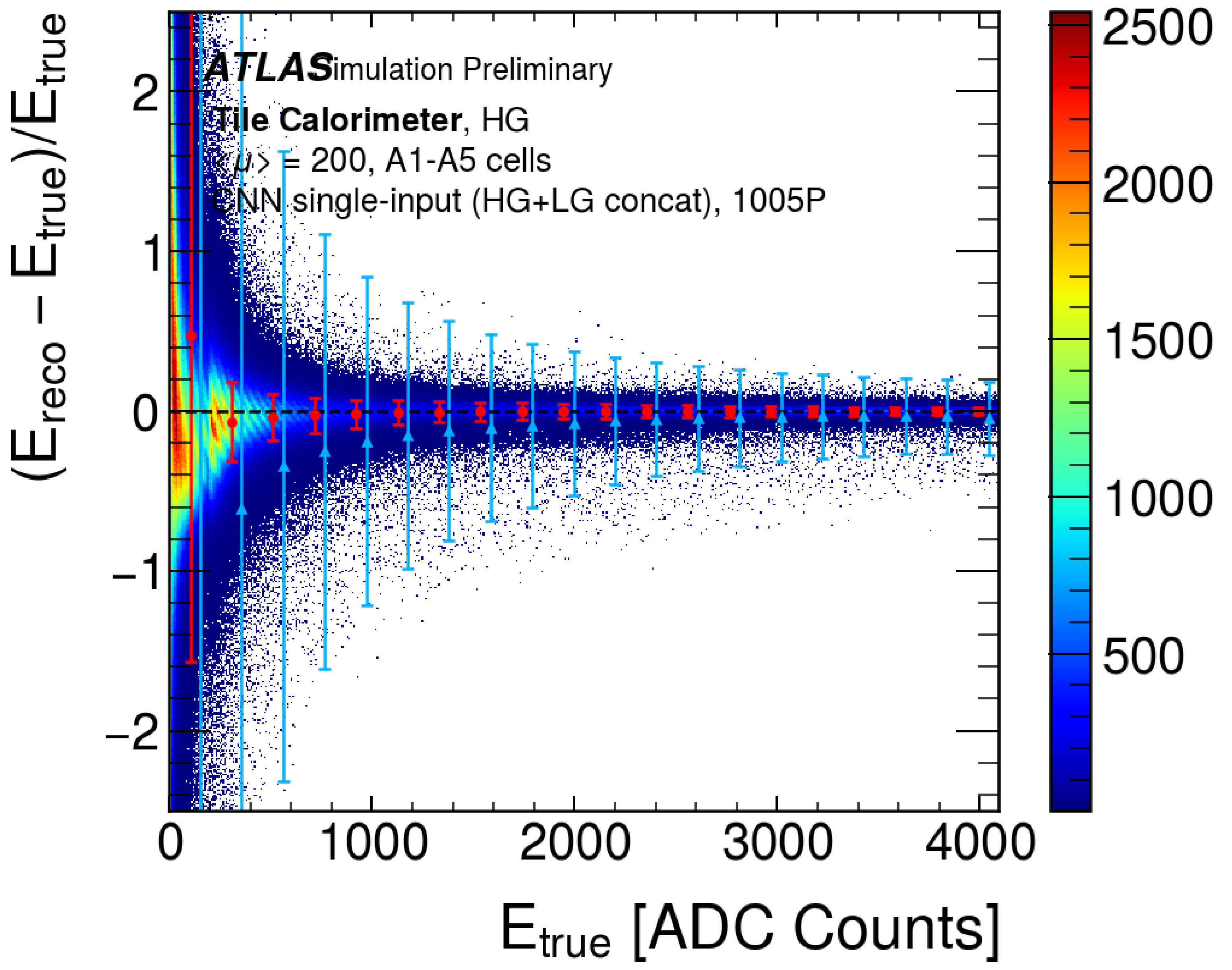


Teacher (1005P), linearity, HG

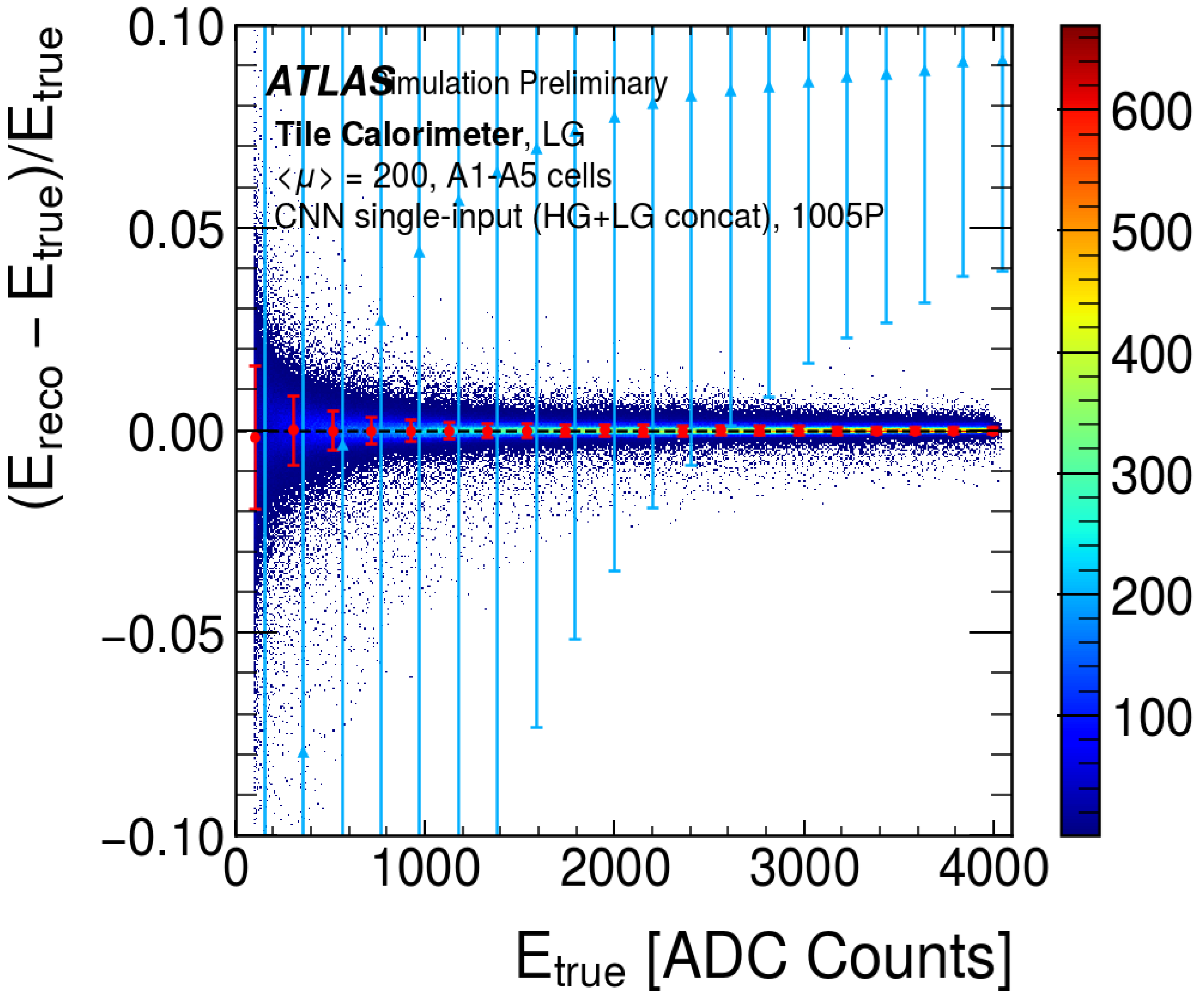


Teacher (1005P), linearity, LG

Backup: Linearity and fractional resolution (teacher vs student)

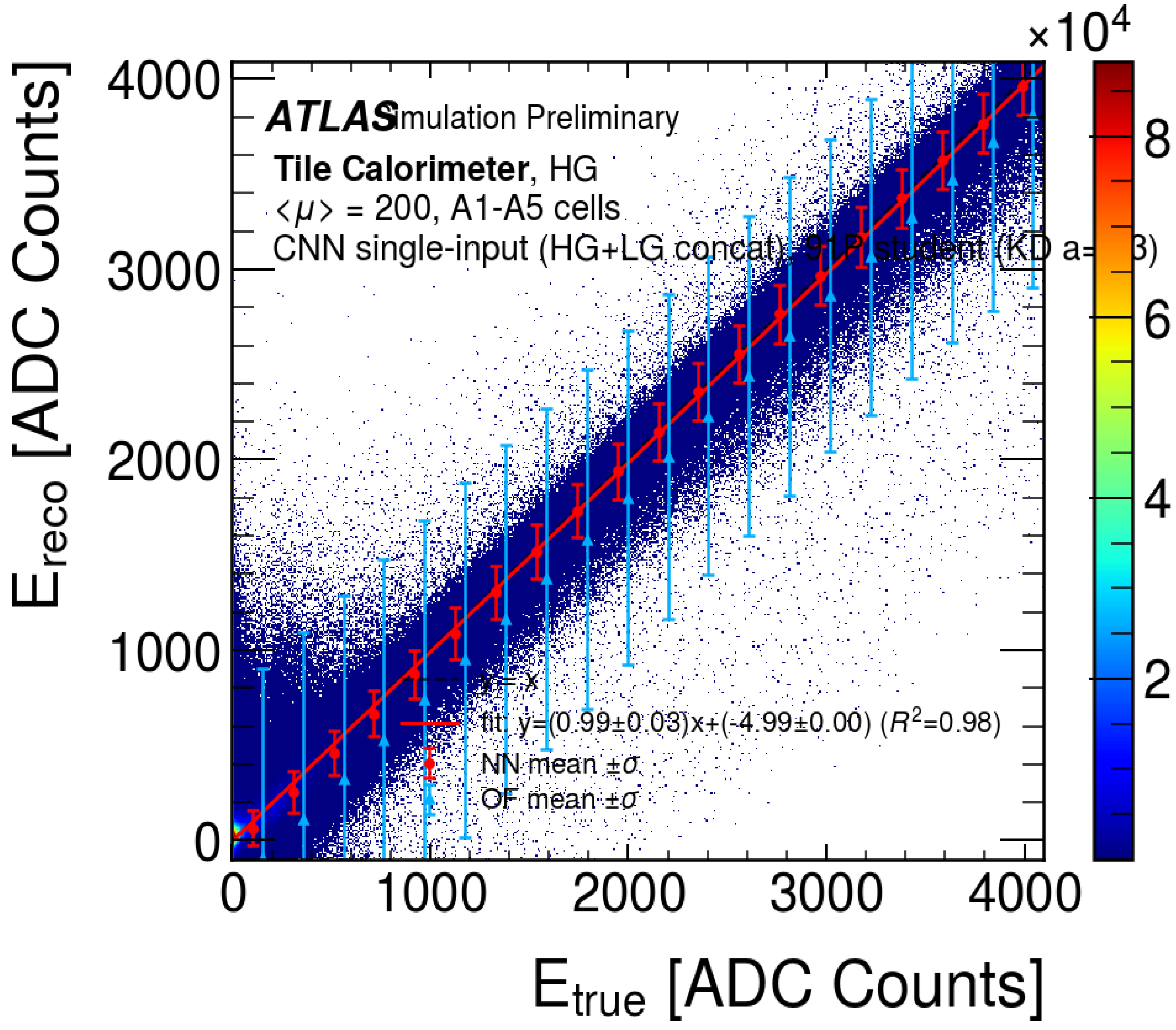


Teacher (1005P), Fractional Resolution, HG

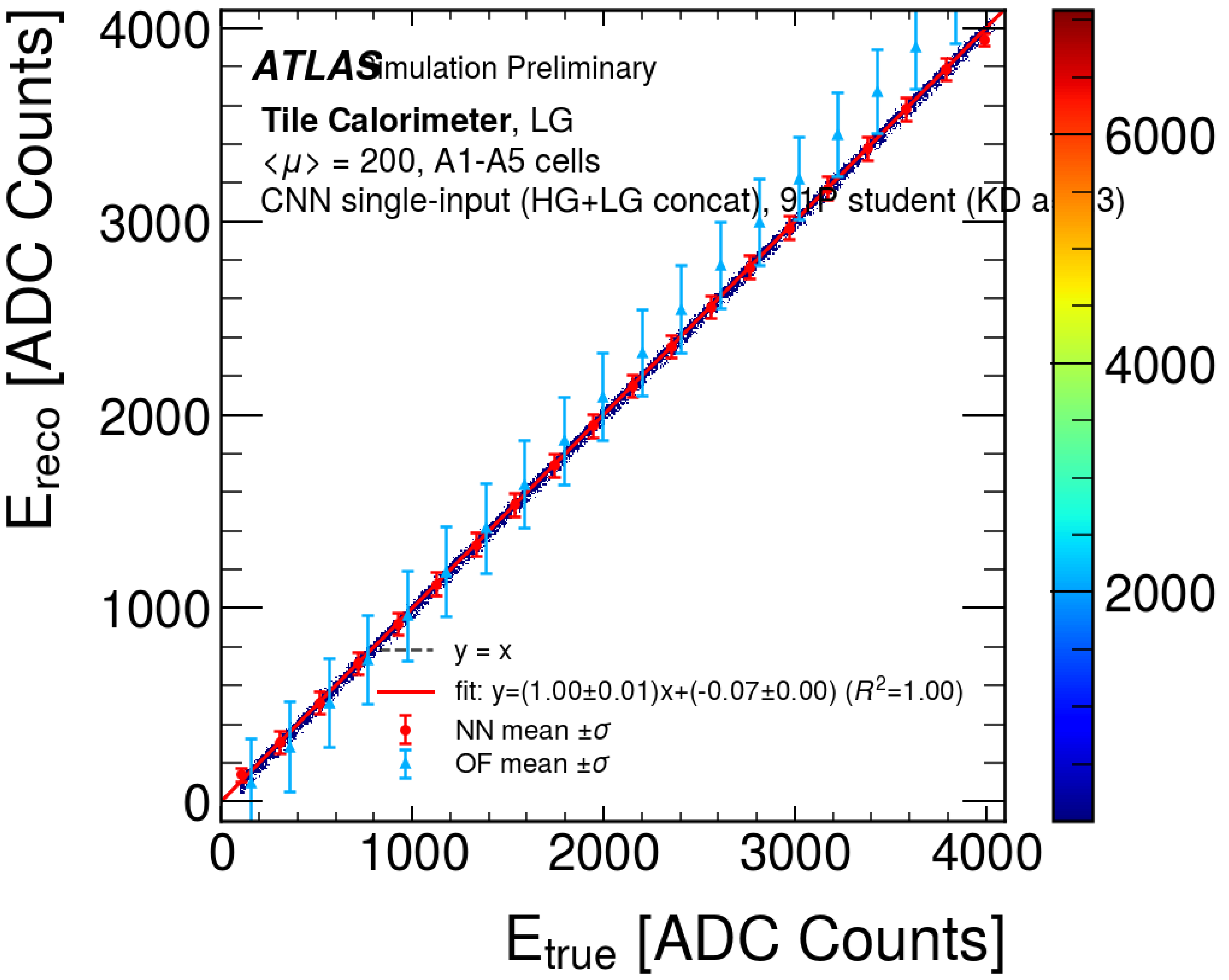


Teacher (1005P), Fractional Resolution, LG

Backup: Linearity and fractional resolution (teacher vs student)

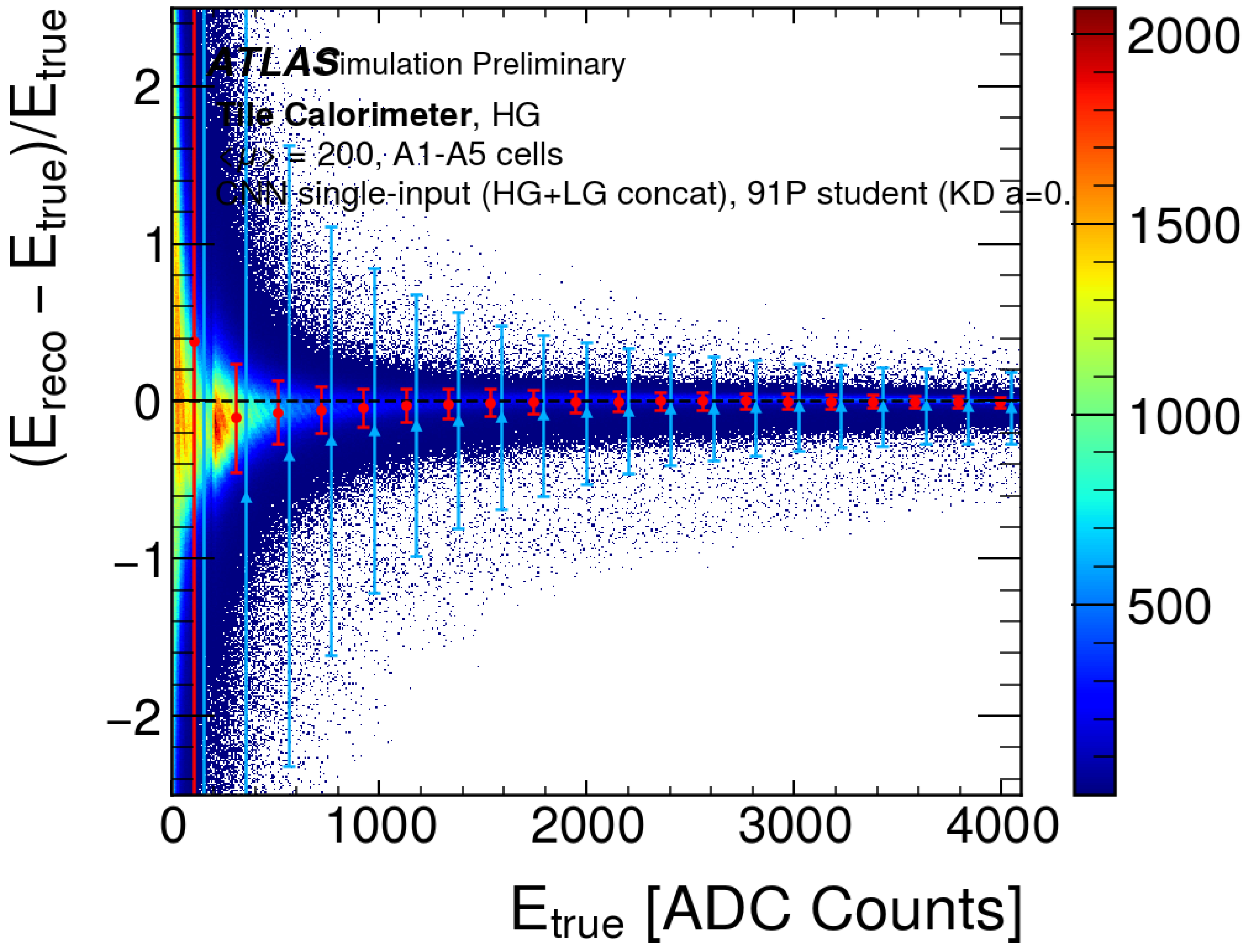


91P student, (KD alpha = 0.3), linearity HG

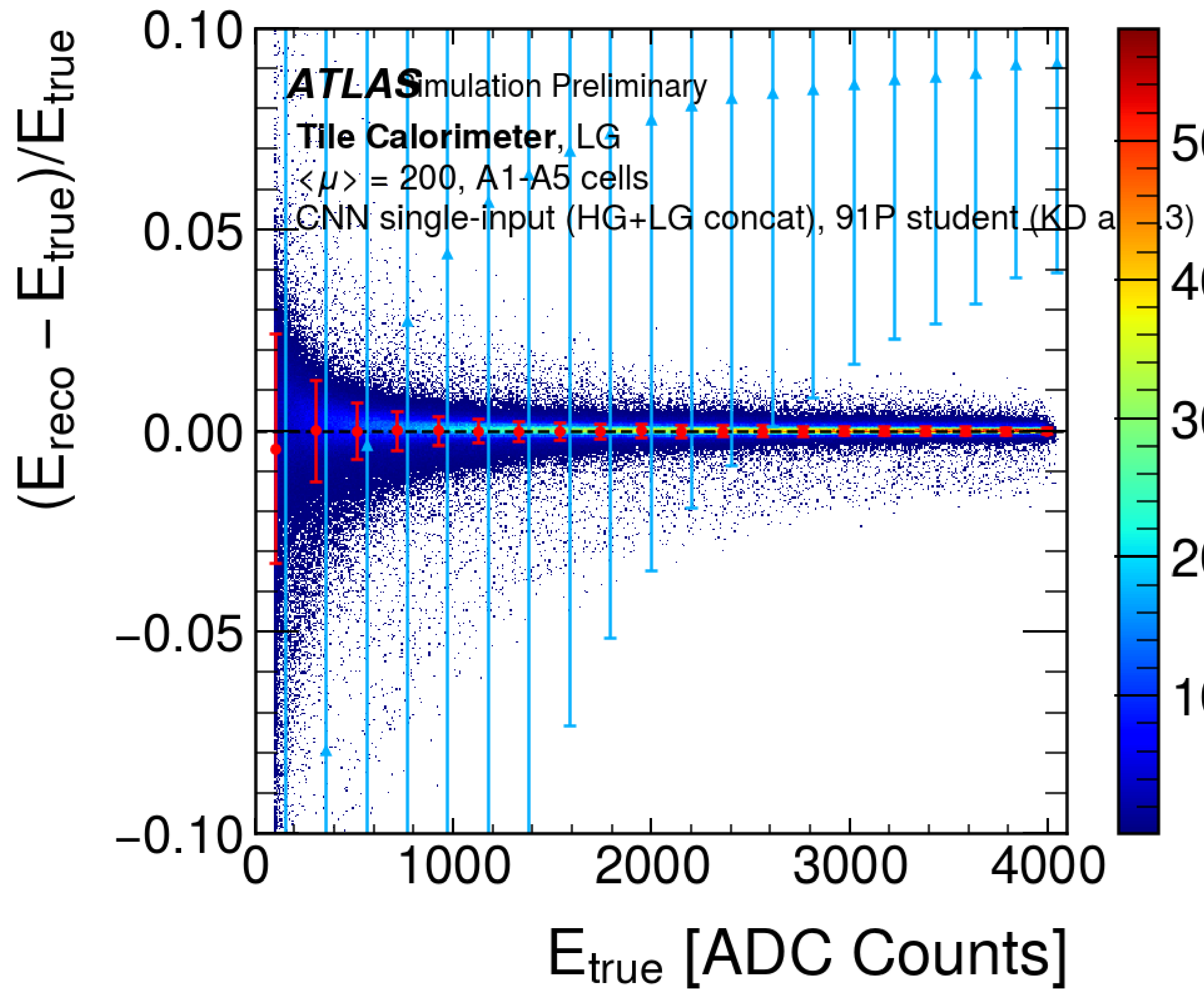


91P student, (KD alpha = 0.3), linearity LG

Backup: Linearity and fractional resolution (teacher vs student)



91P student, (KD alpha = 0.3), Fractional Resolution, HG



91P student, (KD alpha = 0.3), Fractional Resolution, LG

Backup: Architectures and methodology

Teacher

Conv1d(1->18, k5) -> Conv1d(18->14, k3) -> Linear(126->1), LeakyReLU(0.25), **1005 parameters**.

Student (best)

Conv1d(1->4, k5) -> Conv1d(4->3, k3) -> Linear(27->1), LeakyReLU(0.25), **91 parameters**.

Resolution metrics

Global sigma = std over all events (event-weighted). **Bin-mean sigma** = average of the per-energy-bin sigma (energy-flat).

Data filters

Per-sample gain mixing (substitute LG x40 when HG saturates), drop a window if any LG sample saturates (> 4095 ADC) or any sample is below the 10 ADC noise floor, or the central-BC target is out of range.

Backup: Future directions (ML side)

Other cells beyond the A-layer: the BC and D layers, and the E gap scintillators (crack region, a different pulse regime).

Pile-up dependence: how resolution scales with <mu>.

Timing / phase reconstruction as a secondary target.

THANK YOU🙏